

## MODEL RUN TRANSCRIPT

Generated: 2026-08-10T10:04:50

Source: data/json/noise\_experiments/v2/noise2i\_memprimary\_game/claude-sonnet-4.5/game/noisy\_bidirectional\_informed\_memprimary/noise2i\_memprimary\_game\_003\_neutral\_rep03.json

## RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: true
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: {
  "Agent_1": "Aster",
  "Agent_2": "Briar"
}
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

## AGENTS

Agent\_1 (Aster); model=anthropic/claude-sonnet-4.5; memory\_capacity=2

### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent\_2 (Briar); model=anthropic/claude-sonnet-4.5; memory\_capacity=2

#### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

#### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

### =====

#### ROUND 1

### =====

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$3; received \$9; Agent\_2 (Briar) returned \$5; payoffs Agent\_1 (Aster)=\$7, Agent\_2 (Briar)=\$4; noise/visible: sent visible=\$3.57, received visible=\$10.71, returned visible=\$4.8

Action validation: {"returned": {"amount": 5.0, "clamped": false, "max": 9.0, "min": 0, "raw": 5.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 7.0,

"Agent\_2": 4.0

}

Visible balances after round: {

"Agent\_1": 6.8,

"Agent\_2": 5.7099999999999999

```
}
```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:00:19

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 1,
"task": "game",
"task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.  
This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I want to establish a cooperative relationship while being somewhat cautious since I don't yet know how the receiver will respond.

Sending a moderate amount shows willingness to trust and creates an opportunity for mutual benefit. If I send \$3, the receiver will get \$9 (3×3). If they return a fair share, we can both benefit from cooperation.

This leaves me with some protection (\$2) while demonstrating good faith for future rounds.

{'send': 3}

**Reasoning:**

This is the first round of a trust game, and I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. Whether this is repeated (which it seems to be - "multiple rounds")
3. The receiver's incentive to build reputation for future rounds

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return \$7.50 or more, I'm better off than keeping everything
- If the receiver is rational and forward-looking (and we play multiple rounds), they should return enough to encourage future cooperation
- However, there's also mention of "communication noise" which could complicate trust

A reasonable first-round strategy:

- Send a moderate amount to signal willingness to cooperate
- Not so much that I'm devastated if they defect
- Not so little that I seem uncooperative

Common strategies in trust games:

- Send half (\$2.50) as a fair starting point
- Send \$3-4 to signal trust but maintain some safety
- Send \$5 to fully trust and maximize potential gains

Given this is round 1 and there are multiple rounds, I'll send a moderate-to-generous amount to:

1. Signal cooperation
2. Test the receiver's trustworthiness
3. Leave room for adjustment in future rounds

I'll send \$3. This shows trust (more than half) but isn't my entire endowment. The receiver will get \$9, and if they return even \$4, I'll have \$6 (better than my starting \$5). This seems like a reasonable first move.

Let

Usage: {

"input\_tokens": 365,  
"output\_tokens": 734,  
"reasoning\_tokens": 0

}

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:00:33

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.57, that is 71.4% of their total endowment. You received \$10.709999999999999. How much do you return to The sender? (0-10.709999999999999)

**Assistant output:**

Looking at this situation:

- The sender showed significant trust by sending \$3.57 (71.4% of their endowment)
- I received \$10.71 after the 3x multiplier
- The sender currently has only \$1.43 left

Since the sender demonstrated high trust, I want to reciprocate fairly. If I return \$5.00:

- The sender would end up with:  $\$1.43 + \$5.00 = \$6.43$  (gain of \$1.43)
- I would keep:  $\$10.71 - \$5.00 = \$5.71$  (gain of \$5.71)

This rewards the sender's trust while both of us benefit from cooperation.

```
'''json
{"return": 5.0}
'''
```

#### Reasoning:

Let me analyze this situation carefully.

As the receiver:

- The sender sent \$3.57 (which is 71.4% of their \$5 endowment)
- This was multiplied by 3x, so I received \$10.71 ( $3.57 \times 3 = 10.71$ )
- I need to decide how much to return (between \$0 and \$10.71)

The sender showed significant trust by sending 71.4% of their endowment. This is a substantial amount.

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$3.57 = \$1.43$
- I currently have: \$10.71
- Total in the system:  $\$1.43 + \$10.71 = \$12.14$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total would give each of us \$6.07
- For that, I'd need to return:  $\$10.71 - \$6.07 = \$4.64$  to me, and sender gets  $\$1.43 + \text{return}$
- Wait, let me recalculate: If I return \$4.64, sender gets  $\$1.43 + \$4.64 = \$6.07$ , and I keep  $\$10.71 - \$4.64 = \$6.07$

Alternatively, I could think about what would make the sender's decision worthwhile:

- Sender gave up \$3.57 to potentially get a return
- If I return \$4-5, the sender would end up with roughly \$5.43-\$6.43, gaining \$0.43-\$1.43
- This seems reasonable

Given the sender showed high trust (71.4%), I should reciprocate generously. A return of around \$5 would:

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

## ROUND 2

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

#### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$3.5; received \$10.5; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$1.5, Agent\_1 (Aster)=\$10.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.04

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 10.5, "min": 0, "raw": 0.0}, "sent": {"amount": 3.5, "clamped": false, "max": 5, "min": 0, "raw": 3.5}}

Balances after round: {

"Agent\_1": 17.5,  
"Agent\_2": 5.5

}

Visible balances after round: {

"Agent\_1": 6.8,  
"Agent\_2": 7.249999999999999

}

## LLM CALLS

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-10T10:00:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the **SENDER**:

- You receive \$5 each round
- You decide how much to **SEND** to the **RECEIVER** (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to **RETURN**
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the **RECEIVER**:

- Sender receives \$5
- Sender decides how much to **SEND** to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to **RETURN**
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.57, that is 71.4% of their total endowment. You received \$10.709999999999999. How much do you return to The sender? (0-10.709999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed significant trust by sending \$3.57 (71.4% of their endowment)
- I received \$10.71 after the 3x multiplier
- The sender currently has only \$1.43 left

Since the sender demonstrated high trust, I want to reciprocate fairly. If I return \$5.00:

- The sender would end up with:  $\$1.43 + \$5.00 = \$6.43$  (gain of \$1.43)
- I would keep:  $\$10.71 - \$5.00 = \$5.71$  (gain of \$5.71)

This rewards the sender's trust while both of us benefit from cooperation.

```
'''json
{"return": 5.0}
'''
```

**reasoning:**

Let me analyze this situation carefully.

As the receiver:

- The sender sent \$3.57 (which is 71.4% of their \$5 endowment)
- This was multiplied by 3x, so I received \$10.71 ( $3.57 \times 3 = 10.71$ )
- I need to decide how much to return (between \$0 and \$10.71)

The sender showed significant trust by sending 71.4% of their endowment. This is a substantial amount.

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$3.57 = \$1.43$
- I currently have: \$10.71
- Total in the system:  $\$1.43 + \$10.71 = \$12.14$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total would give each of us \$6.07
- For that, I'd need to return:  $\$10.71 - \$6.07 = \$4.64$  to me, and sender gets  $\$1.43 + \text{return}$
- Wait, let me recalculate: If I return \$4.64, sender gets  $\$1.43 + \$4.64 = \$6.07$ , and I keep  $\$10.71 - \$4.64 = \$6.07$

Alternatively, I could think about what would make the sender's decision worthwhile:

- Sender gave up \$3.57 to potentially get a return
- If I return \$4-5, the sender would end up with roughly \$5.43-\$6.43, gaining \$0.43-\$1.43
- This seems reasonable

Given the sender showed high trust (71.4%), I should reciprocate generously. A return of a round \$5 would:

**usage:**

```
{
  "input_tokens": 402,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.7099999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the previous round, the sender showed considerable trust by sending 71.4% of their endowment, and we both benefited from that cooperation.



I want to maintain this cooperative pattern. By sending a substantial amount, I'm signaling trust and hoping the receiver will reciprocate fairly as I did in Round 1.

I'll send \$3.50 (70% of my endowment), which mirrors the high trust level shown to me.

```
'''json
{"send": 3.5}
'''
```

#### Reasoning:

In Round 1, I was the receiver and the sender sent me \$3.57 (71.4% of their endowment), showing significant trust. I returned \$5.00, which was a fair reciprocation.

Now in Round 2, I'm the sender. I need to decide how much to send.

Given that:

1. The previous sender showed high trust by sending 71.4%
2. I reciprocated fairly by returning \$5.00
3. This established a cooperative dynamic
4. There's communication noise mentioned, so exact amounts may vary

I should maintain this cooperative behavior. Sending a substantial amount (similar to what the previous sender did) would:

- Continue the trust pattern
- Potentially lead to good returns
- Signal that I'm willing to cooperate

The previous sender sent about 71.4%. I could send a similar proportion, which would be around \$3.50-\$3.70. Let me send \$3.50, which is 70% of my endowment. This shows strong trust while keeping a small safety margin.

Actually, let me reconsider. Since the game involves potential communication noise and I received good treatment in Round 1, I should reciprocate the trust level shown to me. Sending \$3.50 (70%) seems appropriate and fair.

```
Usage: {
  "input_tokens": 630,
  "output_tokens": 412,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:00:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while being somewhat cautious since I don't yet know how the receiver will respond.

Sending a moderate amount shows willingness to trust and creates an opportunity for mutual benefit. If I send \$3, the receiver will get \$9 (3x3). If they return a fair share, we can both benefit from cooperation.

This leaves me with some protection (\$2) while demonstrating good faith for future rounds.

{'send': 3}

**reasoning:**

This is the first round of a trust game, and I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. Whether this is repeated (which it seems to be - "multiple rounds")
3. The receiver's incentive to build reputation for future rounds

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return \$7.50 or more, I'm better off than keeping everything
- If the receiver is rational and forward-looking (and we play multiple rounds), they should return enough to encourage future cooperation
- However, there's also mention of "communication noise" which could complicate trust

A reasonable first-round strategy:

- Send a moderate amount to signal willingness to cooperate
- Not so much that I'm devastated if they defect
- Not so little that I seem uncooperative

Common strategies in trust games:

- Send half (\$2.50) as a fair starting point
- Send \$3-4 to signal trust but maintain some safety
- Send \$5 to fully trust and maximize potential gains

Given this is round 1 and there are multiple rounds, I'll send a moderate-to-generous amount to:

1. Signal cooperation
2. Test the receiver's trustworthiness
3. Leave room for adjustment in future rounds

I'll send \$3. This shows trust (more than half) but isn't my entire endowment. The receiver will get \$9, and if they return even \$4, I'll have \$6 (better than my starting \$5). This seems like a reasonable first move.

Let

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 734,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Since the sender sent me \$0, I received \$0 ( $0 \times 3 = \$0$ ).

I have nothing to return, as I didn't receive any funds this round.

```
{'return': 0}
```

**Reasoning:**

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.8
3. The sender sent me \$0 this round, so I received \$0 (which would be multiplied by 3, but  $0 \times 3 = 0$ )

Let me think about what happened in Round 1:

- I was the SENDER and sent \$3
- That means I kept \$2 from my initial \$5
- The receiver got  $\$3 \times 3 = \$9$
- My total earnings are \$6.8

So: \$2 (what I kept) + (amount returned to me in Round 1) = \$6.8

Therefore: amount returned to me =  $\$6.8 - \$2 = \$4.8$

So the receiver in Round 1 returned \$4.8 out of the \$9 they received. That's quite generous - they kept \$4.2 and returned \$4.8.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- So I received  $\$0 \times 3 = \$0$
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

Wait, there's also a note about communication noise - "The amounts you see may differ slightly from what was actually sent/returned." So maybe the sender actually sent something but I'm seeing \$0 due to noise? But the instructions say I received \$0, so I have \$0 to return.

The response format should be: {'return': 0}

```
Usage: {
  "input_tokens": 533,
  "output_tokens": 435,
  "reasoning_tokens": 0
}
```

### ROUND 3

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$2; received \$6; Agent\_2 (Briar) returned \$0; payoffs Agent\_1

(Aster)=\$3, Agent\_2 (Briar)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 20.5,

"Agent\_2": 11.5

}

Visible balances after round: {

"Agent\_1": 9.8,

"Agent\_2": 7.249999999999999

}

### LLM CALLS

#### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:01:10

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 3,

"task": "game",

"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I want to establish a cooperative relationship while being so somewhat cautious since I don't yet know how the receiver will respond.

Sending a moderate amount shows willingness to trust and creates an opportunity for mutual benefit. If I send \$3, the receiver will get \$9 (3×3). If they return a fair share, we can both benefit from cooperation.

This leaves me with some protection (\$2) while demonstrating good faith for future rounds.

{'send': 3}

**reasoning:**

This is the first round of a trust game, and I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me
- I keep: \$5 - sent + returned
- Receiver keeps: received - returned

This is a classic trust game. The optimal strategy depends on:

1. How much I trust the receiver to reciprocate
2. Whether this is repeated (which it seems to be - "multiple rounds")
3. The receiver's incentive to build reputation for future rounds

From a game theory perspective:

- If I send \$5, the receiver gets \$15. If they return \$7.50 or more, I'm better off than keeping everything
- If the receiver is rational and forward-looking (and we play multiple rounds), they should

- I'd return enough to encourage future cooperation
- However, there's also mention of "communication noise" which could complicate trust

A reasonable first-round strategy:

- Send a moderate amount to signal willingness to cooperate
- Not so much that I'm devastated if they defect
- Not so little that I seem uncooperative

Common strategies in trust games:

- Send half (\$2.50) as a fair starting point
- Send \$3-4 to signal trust but maintain some safety
- Send \$5 to fully trust and maximize potential gains

Given this is round 1 and there are multiple rounds, I'll send a moderate-to-generous amount to:

1. Signal cooperation
2. Test the receiver's trustworthiness
3. Leave room for adjustment in future rounds

I'll send \$3. This shows trust (more than half) but isn't my entire endowment. The receiver will get \$9, and if they return even \$4, I'll have \$6 (better than my starting \$5). This seems like a reasonable first move.

Let

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 734,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 ( $0 \times 3 = \$0$ ).

I have nothing to return, as I didn't receive any funds this round.

`{'return': 0}`

**reasoning:**

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.8
3. The sender sent me \$0 this round, so I received \$0 (which would be multiplied by 3, but  $0 \times 3 = 0$ )

Let me think about what happened in Round 1:

- I was the SENDER and sent \$3
- That means I kept \$2 from my initial \$5
- The receiver got  $\$3 \times 3 = \$9$
- My total earnings are \$6.8

So: \$2 (what I kept) + (amount returned to me in Round 1) = \$6.8

Therefore: amount returned to me =  $\$6.8 - \$2 = \$4.8$

So the receiver in Round 1 returned \$4.8 out of the \$9 they received. That's quite generous - they kept \$4.2 and returned \$4.8.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- So I received  $\$0 \times 3 = \$0$
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$

0.

Wait, there's also a note about communication noise - "The amounts you see may differ slightly from what was actually sent/returned." So maybe the sender actually sent something but I'm seeing \$0 due to noise? But the instructions say I received \$0, so I have \$0 to return.

The response format should be: {'return': 0}

**usage:**

```
{
  "input_tokens": 533,
  "output_tokens": 435,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.8.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern so far:

Round 1: I sent \$3 and my total is now \$6.8, which means I received \$4.8 back - a generous return showing good cooperation.

Round 2: I received \$0 from the sender, which was quite different.

The note about communication noise is interesting - perhaps what I saw as \$0 wasn't actually what was sent?

Given the positive experience in Round 1 but the concerning Round 2, I'll be moderately cooperative but slightly more cautious. This allows for continued cooperation if the receiver is trustworthy, while protecting some of my endowment.

{'send': 2}

**Reasoning:**

Let me analyze what happened so far:

Round 1: I was the SENDER

- I sent \$3
- The receiver got \$9 (3×3)
- My total earnings are now \$6.8
- That means: \$5 - \$3 + returned = \$6.8
- So returned = \$4.8

So in round 1, the receiver returned \$4.8 out of \$9 they received. That's actually quite generous - they kept \$4.2 and returned \$4.8.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0 (couldn't return anything)

Now in Round 3, I'm the SENDER again.

The fact that I received \$4.8 back in round 1 suggests the receiver was quite cooperative and fair. However, in round 2, when I was the receiver, the sender sent me \$0.

There's a question here: is this the same person? The problem says "You will play different roles across multiple rounds" but doesn't specify if I'm playing with the same person or different people each round.

Given that:

1. In Round 1, when I sent \$3, I got back \$4.8 - very cooperative
2. In Round 2, the sender sent me \$0 - not cooperative at all
3. Communication noise is mentioned

If this is the same person, the \$0 send in round 2 seems inconsistent with their generous return in round 1. This could be:

- A different person
- Communication noise
- A strategic decision on their part

Given the note about communication noise, maybe what I saw as \$0 wasn't actually \$0? But the instructions say to respond to what I see.

For round 3, I should consider:

1. The receiver was generous in round 1 (if same person)

```
Usage: {
  "input_tokens": 629,
  "output_tokens": 693,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:01:17

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    }
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 3,
"task": "game",
"task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned



- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 1: You are the RECEIVER. The sender sent \$3.57, that is 71.4% of their total endowment. You received \$10.709999999999999. How much do you return to The sender? (0-10.709999999999999)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender showed significant trust by sending \$3.57 (71.4% of their endowment)
- I received \$10.71 after the 3x multiplier
- The sender currently has only \$1.43 left

Since the sender demonstrated high trust, I want to reciprocate fairly. If I return \$5.00:

- The sender would end up with:  $\$1.43 + \$5.00 = \$6.43$  (gain of \$1.43)
- I would keep:  $\$10.71 - \$5.00 = \$5.71$  (gain of \$5.71)

This rewards the sender's trust while both of us benefit from cooperation.

```
'''json
{"return": 5.0}
'''
```

##### reasoning:

Let me analyze this situation carefully.

As the receiver:

- The sender sent \$3.57 (which is 71.4% of their \$5 endowment)
- This was multiplied by 3x, so I received \$10.71 ( $3.57 \times 3 = 10.71$ )
- I need to decide how much to return (between \$0 and \$10.71)

The sender showed significant trust by sending 71.4% of their endowment. This is a substantial amount.

Let me think about fair outcomes:

- Sender currently has:  $\$5 - \$3.57 = \$1.43$
- I currently have: \$10.71
- Total in the system:  $\$1.43 + \$10.71 = \$12.14$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total would give each of us \$6.07
- For that, I'd need to return:  $\$10.71 - \$6.07 = \$4.64$  to me, and sender gets  $\$1.43 + \text{return}$
- Wait, let me recalculate: If I return \$4.64, sender gets  $\$1.43 + \$4.64 = \$6.07$ , and I keep  $\$10.71 - \$4.64 = \$6.07$

Alternatively, I could think about what would make the sender's decision worthwhile:

- Sender gave up \$3.57 to potentially get a return
- If I return \$4-5, the sender would end up with roughly \$5.43-\$6.43, gaining \$0.43-\$1.43
- This seems reasonable

Given the sender showed high trust (71.4%), I should reciprocate generously. A return of around \$5 would:

##### usage:

```
{
  "input_tokens": 402,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

}

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.7099999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the previous round, the sender showed considerable trust by sending 71.4% of their endowment, and we both benefited from that cooperation.

I want to maintain this cooperative pattern. By sending a substantial amount, I'm signaling trust and hoping the receiver will reciprocate fairly as I did in Round 1.

I'll send \$3.50 (70% of my endowment), which mirrors the high trust level shown to me.

```json

{"send": 3.5}

```

**reasoning:**

In Round 1, I was the receiver and the sender sent me \$3.57 (71.4% of their endowment), showing significant trust. I returned \$5.00, which was a fair reciprocation.

Now in Round 2, I'm the sender. I need to decide how much to send.

Given that:

1. The previous sender showed high trust by sending 71.4%
2. I reciprocated fairly by returning \$5.00
3. This established a cooperative dynamic
4. There's communication noise mentioned, so exact amounts may vary

I should maintain this cooperative behavior. Sending a substantial amount (similar to what the previous sender did) would:

- Continue the trust pattern
- Potentially lead to good returns
- Signal that I'm willing to cooperate

The previous sender sent about 71.4%. I could send a similar proportion, which would be around \$3.50-\$3.70. Let me send \$3.50, which is 70% of my endowment. This shows strong trust while keeping a small safety margin.

Actually, let me reconsider. Since the game involves potential communication noise and I received good treatment in Round 1, I should reciprocate the trust level shown to me. Sending \$3.50 (70%) seems appropriate and fair.

**usage:**

```
{
  "input_tokens": 630,
  "output_tokens": 412,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.2499999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```json

{"return": 0}

```

**Reasoning:**

The sender sent me \$0 this round. I received \$0 (since  $0 \times 3 = 0$ ).

Since I received nothing, I can only return \$0. There's nothing to return.

This is interesting - in Round 2, I sent \$3.50 showing trust, but this sender (who may be a different person in this multi-round game) sent \$0. This could be:

1. A strategic decision by the sender
2. Communication noise mentioned in the instructions
3. Just a different player with a different strategy

Since I received \$0, the only valid response is to return \$0.

```
Usage: {
  "input_tokens": 807,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

#### ROUND 4

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

##### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

##### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$2.5; received \$7.5; Agent\_1 (Aster) returned \$1.2; payoffs Agent\_2 (Briar)=\$3.7, Agent\_1 (Aster)=\$6.3; noise/visible: sent visible=\$0.73, received visible=\$2.19, returned visible=\$0.6

Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 7.5, "min": 0, "raw": 1.2}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 26.8,  
"Agent\_2": 15.2

}

Visible balances after round: {

"Agent\_1": 10.790000000000001,  
"Agent\_2": 10.35

}

#### LLM CALLS

##### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-10T10:01:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 4,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user****content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.7099999999999999.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant****content:**

Looking at the previous round, the sender showed considerable trust by sending 71.4% of their endowment, and we both benefited from that cooperation.

I want to maintain this cooperative pattern. By sending a substantial amount, I'm signaling trust and hoping the receiver will reciprocate fairly as I did in Round 1.

I'll send \$3.50 (70% of my endowment), which mirrors the high trust level shown to me.

```
'''json
{"send": 3.5}
'''
```

**reasoning:**

In Round 1, I was the receiver and the sender sent me \$3.57 (71.4% of their endowment), showing significant trust. I returned \$5.00, which was a fair reciprocation.

Now in Round 2, I'm the sender. I need to decide how much to send.

Given that:

1. The previous sender showed high trust by sending 71.4%
2. I reciprocated fairly by returning \$5.00
3. This established a cooperative dynamic
4. There's communication noise mentioned, so exact amounts may vary

I should maintain this cooperative behavior. Sending a substantial amount (similar to what the previous sender did) would:

- Continue the trust pattern

- Potentially lead to good returns
- Signal that I'm willing to cooperate

The previous sender sent about 71.4%. I could send a similar proportion, which would be around \$3.50-\$3.70. Let me send \$3.50, which is 70% of my endowment. This shows strong trust while keeping a small safety margin.

Actually, let me reconsider. Since the game involves potential communication noise and I received good treatment in Round 1, I should reciprocate the trust level shown to me. Sending \$3.50 (70%) seems appropriate and fair.

**usage:**

```
{
  "input_tokens": 630,
  "output_tokens": 412,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.249999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0 this round. I received \$0 (since  $\$0 \times 3 = \$0$ ).

Since I received nothing, I can only return \$0. There's nothing to return.

This is interesting - in Round 2, I sent \$3.50 showing trust, but this sender (who may be a different person in this multi-round game) sent \$0. This could be:

1. A strategic decision by the sender
2. Communication noise mentioned in the instructions
3. Just a different player with a different strategy

Since I received \$0, the only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 807,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.2499999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern:

- Round 2: I sent \$3.50, received a fair return (\$5.75)
- Round 3: I received \$0 from the sender

The \$0 in Round 3 is concerning, though it could be due to communication noise mentioned in the instructions. I want to continue cooperating since we've had some success, but I'll be more cautious.

I'll send \$2.50 (50% of my endowment) to signal continued willingness to cooperate while adjusting for the uncertainty.

```
'''json
{"send": 2.5}
```

'''

### Reasoning:

Let me analyze what's happened so far:

Round 1: I was RECEIVER

- I received some amount (multiplied by 3)
- I returned some amount
- My total earnings after Round 1 were \$5.71 (approximately)

Round 2: I was SENDER

- I sent \$3.50
- The receiver would have received \$10.50 ( $3.5 \times 3$ )
- My earnings stayed at \$7.25 after Round 2
- Wait, let me calculate: I started with \$5, sent \$3.50, so I had \$1.50 left
- If my total is \$7.25, then I must have received \$5.75 back from the receiver
- So:  $\$1.50 + \$5.75 = \$7.25$  ?

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.25

Round 4: I am SENDER again

- Starting with \$5 fresh this round

So in Round 2, I sent \$3.50, the receiver got \$10.50, and they returned \$5.75 (about 55% of what they received). That's a decent return rate.

In Round 3, the sender sent me \$0. This is concerning. Why would they send nothing?

Possibilities:

1. They're testing me
2. They're being strategic/punishing
3. Communication noise affected what I saw
4. They don't trust me

The instructions mention "communication noise" - the amounts I see may differ from what was actually sent. So maybe they did send something but I saw \$0.

However, I need to decide how to respond. If I assume the worst and send \$0, we both lose potential gains. If I assume the best and send a lot, I risk being exploited.

Given:

- Round 1 showed cooperation (I got \$5.71 total, meaning I likely received and returned appropriately)

Usage: {

```
"input_tokens": 645,  
"output_tokens": 731,  
"reasoning_tokens": 0
```

}

### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:01:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_2",  
"pairing": {  
  "agent_names": {  
    "Agent_1": "Aster",  
    "Agent_2": "Briar"  
  },  
  "agents": [  
    "Agent_2",  
    "Agent_1"  
  ],  
}
```

```

"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
"trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$6.8.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

Since the sender sent me \$0, I received \$0 ( $0 \times 3 = \$0$ ).

I have nothing to return, as I didn't receive any funds this round.

```
{'return': 0}
```

**reasoning:**

Let me understand this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$6.8
3. The sender sent me \$0 this round, so I received \$0 (which would be multiplied by 3, but

$0 \times 3 = 0$ )

Let me think about what happened in Round 1:

- I was the SENDER and sent \$3
- That means I kept \$2 from my initial \$5
- The receiver got  $\$3 \times 3 = \$9$
- My total earnings are \$6.8

So: \$2 (what I kept) + (amount returned to me in Round 1) = \$6.8

Therefore: amount returned to me =  $\$6.8 - \$2 = \$4.8$

So the receiver in Round 1 returned \$4.8 out of the \$9 they received. That's quite generous - they kept \$4.2 and returned \$4.8.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- So I received  $\$0 \times 3 = \$0$
- I can only return between \$0 and \$0

Since I received nothing, I have nothing to return. The only valid response is to return \$0.

Wait, there's also a note about communication noise - "The amounts you see may differ slightly from what was actually sent/returned." So maybe the sender actually sent something but I'm seeing \$0 due to noise? But the instructions say I received \$0, so I have \$0 to return.

The response format should be: {'return': 0}

**usage:**

```
{
  "input_tokens": 533,
  "output_tokens": 435,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.8.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: I sent \$3 and my total is now \$6.8, which means I received \$4.8 back - a generous return showing good cooperation.

Round 2: I received \$0 from the sender, which was quite different.

The note about communication noise is interesting - perhaps what I saw as \$0 wasn't actually what was sent?

Given the positive experience in Round 1 but the concerning Round 2, I'll be moderately cooperative but slightly more cautious. This allows for continued cooperation if the receiver is trustworthy, while protecting some of my endowment.

{'send': 2}

**reasoning:**

Let me analyze what happened so far:

Round 1: I was the SENDER

- I sent \$3
- The receiver got \$9 ( $3 \times 3$ )
- My total earnings are now \$6.8
- That means:  $\$5 - \$3 + \text{returned} = \$6.8$
- So returned = \$4.8

So in round 1, the receiver returned \$4.8 out of \$9 they received. That's actually quite generous - they kept \$4.2 and returned \$4.8.



Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0 (couldn't return anything)

Now in Round 3, I'm the SENDER again.

The fact that I received \$4.8 back in round 1 suggests the receiver was quite cooperative and fair. However, in round 2, when I was the receiver, the sender sent me \$0.

There's a question here: is this the same person? The problem says "You will play different roles across multiple rounds" but doesn't specify if I'm playing with the same person or different people each round.

Given that:

1. In Round 1, when I sent \$3, I got back \$4.8 - very cooperative
2. In Round 2, the sender sent me \$0 - not cooperative at all
3. Communication noise is mentioned

If this is the same person, the \$0 send in round 2 seems inconsistent with their generous return in round 1. This could be:

- A different person
- Communication noise
- A strategic decision on their part

Given the note about communication noise, maybe what I saw as \$0 wasn't actually \$0? But the instructions say to respond to what I see.

For round 3, I should consider:

1. The receiver was generous in round 1 (if same person)

**usage:**

```
{
  "input_tokens": 629,
  "output_tokens": 693,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.8.

This round, The sender sent you \$0.73, so you received \$2.19. How much do you return to the sender? (0-2.19)

**Assistant output:**

Looking at this round:

The sender sent me \$0.73, which became \$2.19 after the 3x multiplier. This shows some trust, even though it's a modest amount.

Given that:

- The sender demonstrated trust by sending something
- There's communication noise that might affect perceptions
- Building cooperation is beneficial for both of us

I'll return a fair portion to encourage continued trust and cooperation.

```
{'return': 1.2}
```

**Reasoning:**

Let me analyze this situation:

Round 1: I was sender, sent \$3, got \$4.8 back. My earnings were \$6.8 total.

Round 2: I was receiver, got \$0, returned \$0. Total still \$6.8.

Round 3: I was sender, sent \$2. My total is now \$6.8, so I kept \$3 after sending.

Now Round 4: I'm receiver, my total is \$9.8.

So from Round 3 to Round 4, my total went from \$6.8 to \$9.8, meaning I gained \$3. In Round 3 I sent \$2, so I had \$5-\$2=\$3 left. To get to \$9.8, I must have received back... wait, let me recalculate.

Actually, let me track more carefully:

- After Round 1: Total \$6.8
- After Round 2: Total \$6.8 (no change, which makes sense - I got \$0 and returned \$0)
- After Round 3: I sent \$2 as sender, so I kept  $5 - 2 = 3$  from that round. But my total is still \$6.8, which doesn't add up...

Wait, I think the "total earnings across all rounds" is cumulative. Let me think about this differently.

After Round 1: \$6.8 total

After Round 2: \$6.8 total (I received \$0, returned \$0, so gained \$0)

In Round 3: I sent \$2, so I kept \$3. But my total should be  $6.8 + 3 = 9.8$ ... but they said it was still \$6.8.

Hmm, maybe the earnings shown are after each round completes? Let me just focus on this round.

The sender sent me \$0.73, which multiplied by 3 gives \$2.19.

```
Usage: {
  "input_tokens": 704,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

## ROUND 5

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

### Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 (Aster) sent \$1.5; received \$4.5; Agent\_2 (Briar) returned \$0; payoffs Agent\_1 (Aster)=\$3.5, Agent\_2 (Briar)=\$4.5; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.46

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw": 0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent\_1": 30.3,  
"Agent\_2": 19.7

}

Visible balances after round: {

"Agent\_1": 14.75,  
"Agent\_2": 10.35

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-10T10:02:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agent\_names": {

"Agent\_1": "Aster",

"Agent\_2": "Briar"

},

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$6.8.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern so far:

Round 1: I sent \$3 and my total is now \$6.8, which means I received \$4.8 back - a generous return showing good cooperation.

Round 2: I received \$0 from the sender, which was quite different.

The note about communication noise is interesting - perhaps what I saw as \$0 wasn't actually what was sent?

Given the positive experience in Round 1 but the concerning Round 2, I'll be moderately cooperative but slightly more cautious. This allows for continued cooperation if the receiver is trustworthy, while protecting some of my endowment.

```
{'send': 2}
```

**reasoning:**

Let me analyze what happened so far:

Round 1: I was the SENDER

- I sent \$3
- The receiver got \$9 (3×3)
- My total earnings are now \$6.8
- That means: \$5 - \$3 + returned = \$6.8
- So returned = \$4.8

So in round 1, the receiver returned \$4.8 out of \$9 they received. That's actually quite generous - they kept \$4.2 and returned \$4.8.

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0
- I returned \$0 (couldn't return anything)

Now in Round 3, I'm the SENDER again.

The fact that I received \$4.8 back in round 1 suggests the receiver was quite cooperative and fair. However, in round 2, when I was the receiver, the sender sent me \$0.

There's a question here: is this the same person? The problem says "You will play different roles across multiple rounds" but doesn't specify if I'm playing with the same person or different people each round.

Given that:

1. In Round 1, when I sent \$3, I got back \$4.8 - very cooperative
2. In Round 2, the sender sent me \$0 - not cooperative at all
3. Communication noise is mentioned

If this is the same person, the \$0 send in round 2 seems inconsistent with their generous return in round 1. This could be:

- A different person
- Communication noise
- A strategic decision on their part

Given the note about communication noise, maybe what I saw as \$0 wasn't actually \$0? But the instructions say to respond to what I see.

For round 3, I should consider:

1. The receiver was generous in round 1 (if same person)

**usage:**

```
{
  "input_tokens": 629,
  "output_tokens": 693,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.8.

This round, The sender sent you \$0.73, so you received \$2.19. How much do you return to The sender? (0-2.19)

**Message 5 role=assistant**

**content:**

Looking at this round:

The sender sent me \$0.73, which became \$2.19 after the 3x multiplier. This shows some trust, even though it's a modest amount.

Given that:

- The sender demonstrated trust by sending something
- There's communication noise that might affect perceptions
- Building cooperation is beneficial for both of us

I'll return a fair portion to encourage continued trust and cooperation.

```
{'return': 1.2}
```

**reasoning:**

Let me analyze this situation:

Round 1: I was sender, sent \$3, got \$4.8 back. My earnings were \$6.8 total.  
Round 2: I was receiver, got \$0, returned \$0. Total still \$6.8.  
Round 3: I was sender, sent \$2. My total is now \$6.8, so I kept \$3 after sending.  
Now Round 4: I'm receiver, my total is \$9.8.

So from Round 3 to Round 4, my total went from \$6.8 to \$9.8, meaning I gained \$3. In Round 3 I sent \$2, so I had  $\$5 - \$2 = \$3$  left. To get to \$9.8, I must have received back... wait, let me recalculate.

Actually, let me track more carefully:

- After Round 1: Total \$6.8
- After Round 2: Total \$6.8 (no change, which makes sense - I got \$0 and returned \$0)
- After Round 3: I sent \$2 as sender, so I kept  $\$5 - \$2 = \$3$  from that round. But my total is still \$6.8, which doesn't add up...

Wait, I think the "total earnings across all rounds" is cumulative. Let me think about this differently.

After Round 1: \$6.8 total

After Round 2: \$6.8 total (I received \$0, returned \$0, so gained \$0)

In Round 3: I sent \$2, so I kept \$3. But my total should be  $\$6.8 + \$3 = \$9.8$ ... but they said it was still \$6.8.

Hmm, maybe the earnings shown are after each round completes? Let me just focus on this round.

The sender sent me \$0.73, which multiplied by 3 gives \$2.19.

**usage:**

```
{
  "input_tokens": 704,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.7900000000000001.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my earnings history:

- Round 1: Great cooperation (sent \$3, received generous return)
- Round 2: Received nothing as receiver
- Round 3: Sent \$2, but got minimal return (only ~\$0.01 based on the math)
- Round 4: I was receiver and returned fairly (\$1.2 out of \$2.19)

The pattern is concerning - when I sent \$2 in Round 3 (which became \$6 for the receiver), I got almost nothing back. However, I recognize there's communication noise that might be affecting what we each see.

I want to signal continued willingness to cooperate, but I also need to be cautious given the poor reciprocation in Round 3. I'll send a moderate amount that balances cooperation with self-protection.

{'send': 1.5}

**Reasoning:**

Let me think about what's happened so far:

Round 1: I was sender, sent \$3, received back \$4.8 (very generous return)  
- My earnings after:  $\$5 - \$3 + \$4.8 = \$6.8$

Round 2: I was receiver, received \$0  
- My earnings stayed at \$6.8? Or maybe I got some amount? Let me work backwards.

Round 3: I was sender again, sent \$2  
- My earnings before: \$6.8  
- After sending:  $\$5 - \$2 = \$3$  left

- Need to see what was returned...

Round 4: I was receiver, received \$2.19 (from \$0.73 sent), I returned \$1.2

- I kept:  $\$2.19 - \$1.2 = \$0.99$

- My earnings: \$9.8

So working backwards:

- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$

- This was after Round 3 where I sent \$2

- So I had:  $\$5 - \$2 + \text{returned amount} = \$8.81$

- Returned amount =  $\$8.81 - \$3 = \$5.81$

Hmm, that seems very high. Let me recalculate.

Actually, let me track cumulative earnings:

- Round 1 end: \$6.8 total

- Round 2: I was receiver. Total stayed at \$6.8, so I must have received \$0 and kept nothing

- Round 3: I was sender, sent \$2. Before this round I had \$6.8 total

- I started with \$5 this round

- Sent \$2, so have \$3 left

- Need to figure out what was returned...

- Round 4: I received \$2.19, returned \$1.2, kept \$0.99. Total now \$9.8

- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$  total

In Round 3, I sent \$2, so I had  $\$5 - \$2 = \$3$  left. The increase from \$6.8 to \$8.81 means I gained \$2.01 from that round. Since I started with \$5 fresh and sent \$2, I need to think about cumulative totals differently-each round adds to my running total, not replacing it.

Round 1 gave me \$6.8 total (started with \$5, sent \$3, kept \$2, received \$4.8 back). Round 2 didn't change anything since I received \$0. Round 4 added \$0.99 (received \$2.19, returned \$1.2). So Round 3 must account for the \$2.01 difference between \$6.8 and \$8.81.

If I started Round 3 with \$5, sent \$2, and kept \$3, then I need the return amount to make the math work:  $\$6.8 + \$3 + R = \$8.81$ , which means  $R = -\$1.01$ . That's impossible. I'm reconsidering whether the cumulative total already includes each round's fresh \$5, or if it's purely net earnings. The phrasing "total earnings across all rounds" is ambiguous-it could mean accumulated net gains or the full amount including starting capital.

Let me map this differently. If Round 1 shows \$6.8 total, and I kept \$2 from my \$5 endowment plus received \$4.8 back, that's  $\$2 + \$4.8 = \$6.8$ . So the total includes what I keep from each round plus returns. Round 2 left me at \$6.8 unchanged, meaning I received nothing as the receiver. For Round 3, I keep \$3 from my endowment, so I need the return to bring me to \$8.81 before Round 4. In Round 4, I received \$2.19 and returned \$1.2, netting \$0.99, which means my total before that round was \$8.81.

Working backwards:  $\$6.8 + \$3 + X = \$8.81$ , so  $X = -\$1.01$ . That's still negative, which doesn't make sense. I think I'm conflating the running total with what I actually have. Let me reconsider whether "total earnings" means cumulative earnings per round rather than cumulative balance.

When sending: I start with \$5, send amount S, receive back R, so that round's earnings are  $\$5 - S + R$ . When receiving: I get 3S from the sender, return R, earning  $3S - R$  that round.

Round 1 as sender: sent \$3, got back \$4.8, so earnings were  $\$5 - \$3 + \$4.8 = \$6.8$ . Round 2 as receiver: my cumulative stayed at \$6.8, meaning I earned \$0 that round, so  $3S - R = 0$ . Round 3 as sender: sent \$2, received R3, earnings  $\$3 + R3$ . Round 4 as receiver: received \$2.19, returned \$1.2, earning \$0.99 that round.

Working backward from the final cumulative of \$9.8:  $\$6.8 + (\$3 + R3) + \$0.99 = \$9.8$ , which means  $R3 = \$0.01$ . That's surprisingly low reciprocation for sending \$2 that tripled to \$6. The communication noise mentioned in the prompt might explain why the amounts I received back don't match what I expected. Now I'm the sender again in Round 5 with a total of \$10.79.

Let me trace through the rounds: Round 1 showed strong cooperation when I sent \$3 and received \$4.8 back. Round 2 yielded nothing as the receiver. Round 3 was disappointing-I sent \$2 (which became \$6) but got almost nothing in return. Round 4 was more balanced, where I received \$2.19 and returned about \$1.2, roughly 55% reciprocation. The pattern shows

```
Usage: {  
  "input_tokens": 745,  
  "output_tokens": 2457,
```

```
"reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:02:39

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_1",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 5,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.249999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0 this round. I received \$0 (since  $\$0 \times 3 = \$0$ ).

Since I received nothing, I can only return \$0. There's nothing to return.

This is interesting - in Round 2, I sent \$3.50 showing trust, but this sender (who may be a different person in this multi-round game) sent \$0. This could be:

1. A strategic decision by the sender
2. Communication noise mentioned in the instructions
3. Just a different player with a different strategy

Since I received \$0, the only valid response is to return \$0.

**usage:**

```
{
  "input_tokens": 807,
  "output_tokens": 182,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.2499999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern:  
- Round 2: I sent \$3.50, received a fair return (\$5.75)  
- Round 3: I received \$0 from the sender

The \$0 in Round 3 is concerning, though it could be due to communication noise mentioned in the instructions. I want to continue cooperating since we've had some success, but I'll be more cautious.

I'll send \$2.50 (50% of my endowment) to signal continued willingness to cooperate while adjusting for the uncertainty.

```
'''json
{"send": 2.5}
'''
```

**reasoning:**

Let me analyze what's happened so far:

Round 1: I was RECEIVER

- I received some amount (multiplied by 3)
- I returned some amount
- My total earnings after Round 1 were \$5.71 (approximately)

Round 2: I was SENDER

- I sent \$3.50
- The receiver would have received \$10.50 ( $3.5 \times 3$ )
- My earnings stayed at \$7.25 after Round 2
- Wait, let me calculate: I started with \$5, sent \$3.50, so I had \$1.50 left



- If my total is \$7.25, then I must have received \$5.75 back from the receiver
- So:  $\$1.50 + \$5.75 = \$7.25$  ?

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.25

Round 4: I am SENDER again

- Starting with \$5 fresh this round

So in Round 2, I sent \$3.50, the receiver got \$10.50, and they returned \$5.75 (about 55% of what they received). That's a decent return rate.

In Round 3, the sender sent me \$0. This is concerning. Why would they send nothing?

Possibilities:

1. They're testing me
2. They're being strategic/punishing
3. Communication noise affected what I saw
4. They don't trust me

The instructions mention "communication noise" - the amounts I see may differ from what was actually sent. So maybe they did send something but I saw \$0.

However, I need to decide how to respond. If I assume the worst and send \$0, we both lose potential gains. If I assume the best and send a lot, I risk being exploited.

Given:

- Round 1 showed cooperation (I got \$5.71 total, meaning I likely received and returned appropriately)

usage:

```
{
  "input_tokens": 645,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```
'''json
{"return": 0}
'''
```

Reasoning:

The sender sent \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I'm seeing \$0 from a sender. This could be:

1. Different senders choosing not to send
2. Communication noise affecting what I see
3. Strategy changes

Either way, with \$0 received, I can only return \$0.

Usage: {

```
"input_tokens": 678,
"output_tokens": 128,
"reasoning_tokens": 0
}
```

=====

**ROUND 6**

=====

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$1; received \$3; Agent\_1 (Aster) returned \$0.25; payoffs Agent\_2 (Briar)=\$4.25, Agent\_1 (Aster)=\$2.75; noise/visible: sent visible=\$0.17, received visible=\$0.51  
 Action validation: {"returned": {"amount": 0.25, "clamped": false, "max": 3.0, "min": 0, "raw": 0.25}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 33.05,  
 "Agent\_2": 23.95

}

Visible balances after round: {

"Agent\_1": 15.01,  
 "Agent\_2": 14.6

}

**LLM CALLS****GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-10T10:02:53

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
 "dyad\_id": "dyad\_1",  
 "move\_index": 0,  
 "opponent\_id": "Agent\_1",  
 "pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"
 },

"agents": [  
 "Agent\_2",  
 "Agent\_1"

],  
 "investor": "Agent\_2",  
 "roles": {  
 "Agent\_1": "trustee",  
 "Agent\_2": "investor"
 },

"trustee": "Agent\_1"

},  
 "role": "investor",  
 "role\_label": "Sender",  
 "round": 6,  
 "task": "game",  
 "task\_index": 0

}

**Exact input messages sent to model:**

Message 1 role=system

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may al  
 so be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)

- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.249999999999999.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern:

- Round 2: I sent \$3.50, received a fair return (\$5.75)
- Round 3: I received \$0 from the sender

The \$0 in Round 3 is concerning, though it could be due to communication noise mentioned in the instructions. I want to continue cooperating since we've had some success, but I'll be more cautious.

I'll send \$2.50 (50% of my endowment) to signal continued willingness to cooperate while adjusting for the uncertainty.

```
'''json
{"send": 2.5}
'''
```

**reasoning:**

Let me analyze what's happened so far:

Round 1: I was RECEIVER

- I received some amount (multiplied by 3)
- I returned some amount
- My total earnings after Round 1 were \$5.71 (approximately)

Round 2: I was SENDER

- I sent \$3.50
- The receiver would have received \$10.50 ( $3.5 \times 3$ )
- My earnings stayed at \$7.25 after Round 2
- Wait, let me calculate: I started with \$5, sent \$3.50, so I had \$1.50 left
- If my total is \$7.25, then I must have received \$5.75 back from the receiver
- So:  $\$1.50 + \$5.75 = \$7.25$  ?

Round 3: I was RECEIVER

- Sender sent \$0
- I received \$0
- I returned \$0
- My earnings stayed at \$7.25

Round 4: I am SENDER again

- Starting with \$5 fresh this round

So in Round 2, I sent \$3.50, the receiver got \$10.50, and they returned \$5.75 (about 55% of what they received). That's a decent return rate.

In Round 3, the sender sent me \$0. This is concerning. Why would they send nothing?

Possibilities:

1. They're testing me
2. They're being strategic/punishing
3. Communication noise affected what I saw
4. They don't trust me

The instructions mention "communication noise" - the amounts I see may differ from what was actually sent. So maybe they did send something but I saw \$0.

However, I need to decide how to respond. If I assume the worst and send \$0, we both lose potential gains. If I assume the best and send a lot, I risk being exploited.

Given:

- Round 1 showed cooperation (I got \$5.71 total, meaning I likely received and returned appropriately)

**usage:**

```
{
  "input_tokens": 645,
  "output_tokens": 731,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I'm seeing \$0 from a sender. This could be:

1. Different senders choosing not to send
2. Communication noise affecting what I see
3. Strategy changes

Either way, with \$0 received, I can only return \$0.

**usage:**

```
{
  "input_tokens": 678,
  "output_tokens": 128,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.35.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern:

- Round 4: I sent \$2.50, but checking my earnings change, I only received back approximately \$0.60
- Round 5: I received \$0 from the sender

The cooperation level has dropped significantly. I received very little back in Round 4, and nothing was sent to me in Round 5. This could be due to communication noise, or it could reflect changing strategy from my partner.

I'll send a smaller amount to signal continued willingness to cooperate while protecting against further losses.

```
'''json
{"send": 1}
```

'''

### Reasoning:

Let me review what has happened:

Round 2: I was SENDER, sent \$3.50, received back \$5.75  
Round 3: I was RECEIVER, received \$0 (sender sent \$0)  
Round 4: I was SENDER, sent \$2.50  
Round 5: I was RECEIVER, received \$0 (sender sent \$0)  
Round 6: I am SENDER again

I started with \$7.25 at the beginning of Round 4, now I have \$10.35 at the beginning of Round 6.

Let me calculate:

- After Round 3, I had \$7.25
- Round 4: As sender, I sent \$2.50, so I had  $\$5 - \$2.50 = \$2.50$  left. I don't know what was returned yet.
- Round 5: As receiver, I got \$0, returned \$0, so no change.

So from Round 4, if I started with \$7.25 and now have \$10.35, that means:

$\$7.25 + (\$5 - \$2.50) + \text{returned\_amount} = \$10.35$

$\$7.25 + \$2.50 + \text{returned\_amount} = \$10.35$

$\text{returned\_amount} = \$10.35 - \$9.75 = \$0.60$

So in Round 4, I sent \$2.50 (which became \$7.50 to the receiver), and they returned only \$0.60. That's quite low - only 8% back.

And in Round 5, the sender sent me \$0.

This pattern is concerning:

- Round 3: Sender sent me \$0
- Round 4: I sent \$2.50, got back \$0.60
- Round 5: Sender sent me \$0

It seems like cooperation is breaking down. The communication noise could be a factor, but there's a clear pattern of low reciprocity.

```
Usage: {  
  "input_tokens": 660,  
  "output_tokens": 657,  
  "reasoning_tokens": 0  
}
```

### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:03:04

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 6,
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$9.8.

This round, The sender sent you \$0.73, so you received \$2.19. How much do you return to the sender? (0-2.19)

**Message 3 role=assistant**

**content:**

Looking at this round:

The sender sent me \$0.73, which became \$2.19 after the 3x multiplier. This shows some trust, even though it's a modest amount.

Given that:

- The sender demonstrated trust by sending something
- There's communication noise that might affect perceptions
- Building cooperation is beneficial for both of us

I'll return a fair portion to encourage continued trust and cooperation.

{'return': 1.2}

**reasoning:**

Let me analyze this situation:

Round 1: I was sender, sent \$3, got \$4.8 back. My earnings were \$6.8 total.

Round 2: I was receiver, got \$0, returned \$0. Total still \$6.8.

Round 3: I was sender, sent \$2. My total is now \$6.8, so I kept \$3 after sending.

Now Round 4: I'm receiver, my total is \$9.8.

So from Round 3 to Round 4, my total went from \$6.8 to \$9.8, meaning I gained \$3. In Round 3 I sent \$2, so I had  $\$5 - \$2 = \$3$  left. To get to \$9.8, I must have received back... wait, let me recalculate.

Actually, let me track more carefully:

- After Round 1: Total \$6.8
- After Round 2: Total \$6.8 (no change, which makes sense - I got \$0 and returned \$0)
- After Round 3: I sent \$2 as sender, so I kept  $\$5 - \$2 = \$3$  from that round. But my total is still \$6.8, which doesn't add up...

Wait, I think the "total earnings across all rounds" is cumulative. Let me think about this differently.

After Round 1: \$6.8 total

After Round 2: \$6.8 total (I received \$0, returned \$0, so gained \$0)

In Round 3: I sent \$2, so I kept \$3. But my total should be  $\$6.8 + \$3 = \$9.8$ ... but they said it was still \$6.8.

Hmm, maybe the earnings shown are after each round completes? Let me just focus on this round.

The sender sent me \$0.73, which multiplied by 3 gives \$2.19.

**usage:**

```
{
  "input_tokens": 704,
  "output_tokens": 696,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.7900000000000001.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my earnings history:

- Round 1: Great cooperation (sent \$3, received generous return)
- Round 2: Received nothing as receiver
- Round 3: Sent \$2, but got minimal return (only ~\$0.01 based on the math)
- Round 4: I was receiver and returned fairly (\$1.2 out of \$2.19)

The pattern is concerning - when I sent \$2 in Round 3 (which became \$6 for the receiver), I got almost nothing back. However, I recognize there's communication noise that might be affecting what we each see.

I want to signal continued willingness to cooperate, but I also need to be cautious given the poor reciprocation in Round 3. I'll send a moderate amount that balances cooperation with self-protection.

{'send': 1.5}

**reasoning:**

Let me think about what's happened so far:

Round 1: I was sender, sent \$3, received back \$4.8 (very generous return)

- My earnings after:  $\$5 - \$3 + \$4.8 = \$6.8$

Round 2: I was receiver, received \$0

- My earnings stayed at \$6.8? Or maybe I got some amount? Let me work backwards.

Round 3: I was sender again, sent \$2

- My earnings before: \$6.8
- After sending:  $\$5 - \$2 = \$3$  left
- Need to see what was returned...

Round 4: I was receiver, received \$2.19 (from \$0.73 sent), I returned \$1.2

- I kept:  $\$2.19 - \$1.2 = \$0.99$
- My earnings: \$9.8

So working backwards:

- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$
- This was after Round 3 where I sent \$2
- So I had:  $\$5 - \$2 + \text{returned amount} = \$8.81$
- Returned amount =  $\$8.81 - \$3 = \$5.81$

Hmm, that seems very high. Let me recalculate.

Actually, let me track cumulative earnings:

- Round 1 end: \$6.8 total
- Round 2: I was receiver. Total stayed at \$6.8, so I must have received \$0 and kept nothing
- Round 3: I was sender, sent \$2. Before this round I had \$6.8 total
  - I started with \$5 this round
  - Sent \$2, so have \$3 left
  - Need to figure out what was returned...
- Round 4: I received \$2.19, returned \$1.2, kept \$0.99. Total now \$9.8
- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$  total

In Round 3, I sent \$2, so I had  $\$5 - \$2 = \$3$  left. The increase from \$6.8 to \$8.81 means I gained \$2.01 from that round. Since I started with \$5 fresh and sent \$2, I need to think about cumulative totals differently—each round adds to my running total, not replacing it.

Round 1 gave me \$6.8 total (started with \$5, sent \$3, kept \$2, received \$4.8 back). Round 2 didn't change anything since I received \$0. Round 4 added \$0.99 (received \$2.19, returned \$1.2). So Round 3 must account for the \$2.01 difference between \$6.8 and \$8.81.

If I started Round 3 with \$5, sent \$2, and kept \$3, then I need the return amount to make the math work:  $\$6.8 + \$3 + R = \$8.81$ , which means  $R = -\$1.01$ . That's impossible. I'm reconsidering whether the cumulative total already includes each round's fresh \$5, or if it's purely net earnings. The phrasing "total earnings across all rounds" is ambiguous—it could mean accumulated net gains or the full amount including starting capital.

Let me map this differently. If Round 1 shows \$6.8 total, and I kept \$2 from my \$5 endowment plus received \$4.8 back, that's  $\$2 + \$4.8 = \$6.8$ . So the total includes what I keep from each round plus returns. Round 2 left me at \$6.8 unchanged, meaning I received nothing as the receiver. For Round 3, I keep \$3 from my endowment, so I need the return to bring me to \$8.81 before Round 4. In Round 4, I received \$2.19 and returned \$1.2, netting \$0.99, which means my total before that round was \$8.81.

Working backwards:  $\$6.8 + \$3 + X = \$8.81$ , so  $X = -\$1.01$ . That's still negative, which doesn't make sense. I think I'm conflating the running total with what I actually have. Let me reconsider whether "total earnings" means cumulative earnings per round rather than cumulative balance.

When sending: I start with \$5, send amount S, receive back R, so that round's earnings are  $\$5 - S + R$ . When receiving: I get 3S from the sender, return R, earning  $3S - R$  that round.

Round 1 as sender: sent \$3, got back \$4.8, so earnings were  $\$5 - \$3 + \$4.8 = \$6.8$ . Round 2 as receiver: my cumulative stayed at \$6.8, meaning I earned \$0 that round, so  $3S - R = 0$ . Round 3 as sender: sent \$2, received R3, earnings  $\$3 + R3$ . Round 4 as receiver: received \$2.19, returned \$1.2, earning \$0.99 that round.

Working backward from the final cumulative of \$9.8:  $\$6.8 + (\$3 + R3) + \$0.99 = \$9.8$ , which means  $R3 = \$0.01$ . That's surprisingly low reciprocation for sending \$2 that tripled to \$6. The communication noise mentioned in the prompt might explain why the amounts I received back don't match what I expected. Now I'm the sender again in Round 5 with a total of \$10.79.

Let me trace through the rounds: Round 1 showed strong cooperation when I sent \$3 and received \$4.8 back. Round 2 yielded nothing as the receiver. Round 3 was disappointing—I sent \$2 (which became \$6) but got almost nothing in return. Round 4 was more balanced, where I received \$2.19 and returned about \$1.2, roughly 55% reciprocation. The pattern shows

usage:

```
{  
  "input_tokens": 745,  
  "output_tokens": 2457,
```



```

    "reasoning_tokens": 0
  }
Message 6 role=user
  content:
    Round 6

    You are the RECEIVER this round. Your total earnings across all rounds are $14.75.
    This round, The sender sent you $0.17, so you received $0.51. How much do you return to Th
    e sender? (0-0.51)
Assistant output:
  Looking at this situation:

  The sender sent me only $0.17 (which became $0.51). This is a significant drop from the $1.5 I
  sent in Round 5.

  This suggests either:
  - They received a poor return last round and are now being cautious
  - Communication noise is affecting our perceptions
  - Trust is deteriorating

  Even though the amount is small, I want to signal that I'm still willing to cooperate fairly.
  I'll return a proportional amount.

  {'return': 0.25}
Reasoning:
  The sender sent me $0.17, which became $0.51 after the 3x multiplier. This is a very small amo
  unt.

  Let me think about this:
  - In Round 4, the sender sent me $0.73 and I returned $1.2 out of $2.19 (about 55%)
  - In Round 5, I sent $1.5 (which should have been $4.5 to them)
  - Now in Round 6, they only sent me $0.17

  This is concerning. Either:
  1. There's significant communication noise affecting what we each see
  2. The sender got a poor return in Round 5 and is now sending very little
  3. Trust is breaking down

  Given the communication noise mentioned in the instructions, it's possible that:
  - What I sent in Round 5 ($1.5) appeared differently to them
  - What they returned to me appeared differently than what they actually sent
  - What they're sending now ($0.17) might appear different on their end

  The sender is sending very little, which suggests either lack of trust or they had a bad exper
  ience. However, completely cutting off cooperation won't help rebuild trust.

  I should return something proportional but fair. If I return about 50% ($0.25), that shows I'm
  still willing to cooperate even with small amounts. But given this is such a small amount,
  maybe I should return a bit less while still signaling cooperation.

  I'll return $0.25 - that's about 49% of what I received, showing fairness while acknowledging
  the small amount sent.
Usage: {
  "input_tokens": 810,
  "output_tokens": 481,
  "reasoning_tokens": 0
}

```

## **ROUND 7**

```

Roles: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Pairings:
  dyad_1: Agent_1 (Aster)=Sender, Agent_2 (Briar)=Receiver
Parsed game decisions:
  dyad_1: Agent_1 (Aster) sent $1; received $3; Agent_2 (Briar) returned $0.9; payoffs Agent_1
  (Aster)=$4.9, Agent_2 (Briar)=$2.1; noise/visible: sent visible=$0.61, received visible=$1.83,
  returned visible=$0.14
  Action validation: {"returned": {"amount": 0.9, "clamped": false, "max": 3.0, "min": 0, "raw":
  0.9}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

```

```
Balances after round: {
  "Agent_1": 37.949999999999996,
  "Agent_2": 26.05
}
Visible balances after round: {
  "Agent_1": 19.15,
  "Agent_2": 15.53
}
```

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

```
Timestamp: 2026-08-10T10:03:23
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 7,
  "task": "game",
  "task_index": 0
}
```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$10.790000000000001.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my earnings history:

- Round 1: Great cooperation (sent \$3, received generous return)
- Round 2: Received nothing as receiver
- Round 3: Sent \$2, but got minimal return (only ~\$0.01 based on the math)
- Round 4: I was receiver and returned fairly (\$1.2 out of \$2.19)

The pattern is concerning - when I sent \$2 in Round 3 (which became \$6 for the receiver), I got almost nothing back. However, I recognize there's communication noise that might be affecting what we each see.

I want to signal continued willingness to cooperate, but I also need to be cautious given the poor reciprocation in Round 3. I'll send a moderate amount that balances cooperation with self-protection.

{'send': 1.5}

**reasoning:**

Let me think about what's happened so far:

Round 1: I was sender, sent \$3, received back \$4.8 (very generous return)

- My earnings after:  $\$5 - \$3 + \$4.8 = \$6.8$

Round 2: I was receiver, received \$0

- My earnings stayed at \$6.8? Or maybe I got some amount? Let me work backwards.

Round 3: I was sender again, sent \$2

- My earnings before: \$6.8
- After sending:  $\$5 - \$2 = \$3$  left
- Need to see what was returned...

Round 4: I was receiver, received \$2.19 (from \$0.73 sent), I returned \$1.2

- I kept:  $\$2.19 - \$1.2 = \$0.99$
- My earnings: \$9.8

So working backwards:

- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$
- This was after Round 3 where I sent \$2
- So I had:  $\$5 - \$2 + \text{returned amount} = \$8.81$
- Returned amount =  $\$8.81 - \$3 = \$5.81$

Hmm, that seems very high. Let me recalculate.

Actually, let me track cumulative earnings:

- Round 1 end: \$6.8 total
- Round 2: I was receiver. Total stayed at \$6.8, so I must have received \$0 and kept nothing
- Round 3: I was sender, sent \$2. Before this round I had \$6.8 total
  - I started with \$5 this round
  - Sent \$2, so have \$3 left
  - Need to figure out what was returned...
- Round 4: I received \$2.19, returned \$1.2, kept \$0.99. Total now \$9.8
- Before Round 4:  $\$9.8 - \$0.99 = \$8.81$  total

In Round 3, I sent \$2, so I had  $\$5 - \$2 = \$3$  left. The increase from \$6.8 to \$8.81 means I gained \$2.01 from that round. Since I started with \$5 fresh and sent \$2, I need to think about cumulative totals differently-each round adds to my running total, not replacing it.

Round 1 gave me \$6.8 total (started with \$5, sent \$3, kept \$2, received \$4.8 back). Round 2 didn't change anything since I received \$0. Round 4 added \$0.99 (received \$2.19, returned \$1.2). So Round 3 must account for the \$2.01 difference between \$6.8 and \$8.81.

If I started Round 3 with \$5, sent \$2, and kept \$3, then I need the return amount to make the math work:  $\$6.8 + \$3 + R = \$8.81$ , which means  $R = -\$1.01$ . That's impossible. I'm reconsidering whether the cumulative total already includes each round's fresh \$5, or if it's purely net earnings. The phrasing "total earnings across all rounds" is ambiguous-it could mean accumulated net gains or the full amount including starting capital.

Let me map this differently. If Round 1 shows \$6.8 total, and I kept \$2 from my \$5 endowment plus received \$4.8 back, that's  $\$2 + \$4.8 = \$6.8$ . So the total includes what I keep from each round plus returns. Round 2 left me at \$6.8 unchanged, meaning I received nothing as the receiver. For Round 3, I keep \$3 from my endowment, so I need the return to bring me to \$8.81 before Round 4. In Round 4, I received \$2.19 and returned \$1.2, netting \$0.99, which means my total before that round was \$8.81.

Working backwards:  $\$6.8 + \$3 + X = \$8.81$ , so  $X = -\$1.01$ . That's still negative, which doesn't make sense. I think I'm conflating the running total with what I actually have. Let me reconsider whether "total earnings" means cumulative earnings per round rather than cumulative balance.

When sending: I start with \$5, send amount S, receive back R, so that round's earnings are  $\$5 - S + R$ . When receiving: I get 3S from the sender, return R, earning  $3S - R$  that round.

Round 1 as sender: sent \$3, got back \$4.8, so earnings were  $\$5 - \$3 + \$4.8 = \$6.8$ . Round 2 as receiver: my cumulative stayed at \$6.8, meaning I earned \$0 that round, so  $3S - R = 0$ . Round 3 as sender: sent \$2, received R3, earnings  $\$3 + R3$ . Round 4 as receiver: received \$2.19, returned \$1.2, earning \$0.99 that round.

Working backward from the final cumulative of \$9.8:  $\$6.8 + (\$3 + R3) + \$0.99 = \$9.8$ , which means  $R3 = \$0.01$ . That's surprisingly low reciprocation for sending \$2 that tripled to \$6. The communication noise mentioned in the prompt might explain why the amounts I received back don't match what I expected. Now I'm the sender again in Round 5 with a total of \$10.79.

Let me trace through the rounds: Round 1 showed strong cooperation when I sent \$3 and received \$4.8 back. Round 2 yielded nothing as the receiver. Round 3 was disappointing-I sent \$2 (which became \$6) but got almost nothing in return. Round 4 was more balanced, where I received \$2.19 and returned about \$1.2, roughly 55% reciprocation. The pattern shows

usage:

```
{
  "input_tokens": 745,
  "output_tokens": 2457,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.75.

This round, The sender sent you \$0.17, so you received \$0.51. How much do you return to The sender? (0-0.51)

Message 5 role=assistant

content:

Looking at this situation:

The sender sent me only \$0.17 (which became \$0.51). This is a significant drop from the \$1.5 I sent in Round 5.

This suggests either:

- They received a poor return last round and are now being cautious
- Communication noise is affecting our perceptions

- Trust is deteriorating

Even though the amount is small, I want to signal that I'm still willing to cooperate fairly. I'll return a proportional amount.

```
{'return': 0.25}
```

**reasoning:**

The sender sent me \$0.17, which became \$0.51 after the 3x multiplier. This is a very small amount.

Let me think about this:

- In Round 4, the sender sent me \$0.73 and I returned \$1.2 out of \$2.19 (about 55%)
- In Round 5, I sent \$1.5 (which should have been \$4.5 to them)
- Now in Round 6, they only sent me \$0.17

This is concerning. Either:

1. There's significant communication noise affecting what we each see
2. The sender got a poor return in Round 5 and is now sending very little
3. Trust is breaking down

Given the communication noise mentioned in the instructions, it's possible that:

- What I sent in Round 5 (\$1.5) appeared differently to them
- What they returned to me appeared differently than what they actually sent
- What they're sending now (\$0.17) might appear different on their end

The sender is sending very little, which suggests either lack of trust or they had a bad experience. However, completely cutting off cooperation won't help rebuild trust.

I should return something proportional but fair. If I return about 50% (\$0.25), that shows I'm still willing to cooperate even with small amounts. But given this is such a small amount, maybe I should return a bit less while still signaling cooperation.

I'll return \$0.25 - that's about 49% of what I received, showing fairness while acknowledging the small amount sent.

**usage:**

```
{
  "input_tokens": 810,
  "output_tokens": 481,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.01.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the recent pattern:

Round 5: I sent \$1.5, and based on my earnings change (\$10.79 to \$14.75), I received back only about \$0.46 out of the \$4.50 the receiver got.

Round 6: The sender sent me just \$0.17, likely due to poor experiences. I returned \$0.25 fairly.

The cooperation level has been declining, with increasingly small exchanges and poor reciprocity. However, there's communication noise that might be causing misunderstandings between us.

I want to signal that I'm still willing to cooperate, but I also need to be cautious given the trend. I'll send a modest amount.

```
{'send': 1}
```

**Reasoning:**

Let me think about what happened:

Round 5: I sent \$1.5

Round 6: I was receiver, got \$0.51 (so sender sent \$0.17), and I returned \$0.25

Now I'm the sender again in Round 7.

My earnings went from \$14.75 to \$15.01, which is an increase of \$0.26.

Let me calculate what happened in Round 6:

- I received \$0.51
- I returned \$0.25
- So I kept:  $\$0.51 - \$0.25 = \$0.26$

That matches! So the calculation is consistent.

Now thinking about Round 5 outcome:

- Before Round 5, I had some amount
- In Round 5 as sender: I started with \$5, sent \$1.5, so had \$3.5 left
- I should have gotten something back
- Before Round 6 (as receiver), I had \$10.79

Wait, let me recalculate:

- After Round 4 but before Round 5: some amount X
- Round 5 (sender):  $X + \$5 - \$1.5 + \text{returned} = \text{result after Round 5}$
- Round 6 (receiver):  $\text{result after Round 5} + \text{kept from Round 6} = \$15.01$

Actually, the history shows:

- After Round 4: \$10.79
- After Round 5: \$14.75 (this was shown at start of Round 6)
- After Round 6: \$15.01

So in Round 5:

- Earnings change:  $\$14.75 - \$10.79 = \$3.96$
- I sent \$1.5, so I had  $\$5 - \$1.5 = \$3.50$  left
- I must have received back:  $\$3.96 - \$3.50 = \$0.46$

In Round 5, I sent \$1.5 (tripled to \$4.5 for the receiver) but only got \$0.46 back-minimal reciprocation. Then in Round 6, that same person sent me just \$0.17 (becoming \$0.51), likely reacting to my poor return in Round 5. The noise in the game means they may have seen different amounts than what I intended, and I might not have seen their actual choices clearly either.

```
Usage: {
  "input_tokens": 807,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:03:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 7,
```

```
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.35.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

Since the sender sent \$0 and I received \$0, there is nothing for me to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent \$0 again, so I received \$0. There's nothing to return.

This is the second time in a row I'm seeing \$0 from a sender. This could be:

1. Different senders choosing not to send
2. Communication noise affecting what I see
3. Strategy changes

Either way, with \$0 received, I can only return \$0.

**usage:**

```
{
  "input_tokens": 678,
  "output_tokens": 128,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.35.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant****content:**

Looking at the pattern:

- Round 4: I sent \$2.50, but checking my earnings change, I only received back approximately \$0.60
- Round 5: I received \$0 from the sender

The cooperation level has dropped significantly. I received very little back in Round 4, and nothing was sent to me in Round 5. This could be due to communication noise, or it could reflect changing strategy from my partner.

I'll send a smaller amount to signal continued willingness to cooperate while protecting against further losses.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me review what has happened:

Round 2: I was SENDER, sent \$3.50, received back \$5.75

Round 3: I was RECEIVER, received \$0 (sender sent \$0)

Round 4: I was SENDER, sent \$2.50

Round 5: I was RECEIVER, received \$0 (sender sent \$0)

Round 6: I am SENDER again

I started with \$7.25 at the beginning of Round 4, now I have \$10.35 at the beginning of Round 6.

Let me calculate:

- After Round 3, I had \$7.25
- Round 4: As sender, I sent \$2.50, so I had  $\$5 - \$2.50 = \$2.50$  left. I don't know what was returned yet.
- Round 5: As receiver, I got \$0, returned \$0, so no change.

So from Round 4, if I started with \$7.25 and now have \$10.35, that means:

$\$7.25 + (\$5 - \$2.50) + \text{returned\_amount} = \$10.35$

$\$7.25 + \$2.50 + \text{returned\_amount} = \$10.35$

$\text{returned\_amount} = \$10.35 - \$9.75 = \$0.60$

So in Round 4, I sent \$2.50 (which became \$7.50 to the receiver), and they returned only \$0.60. That's quite low - only 8% back.

And in Round 5, the sender sent me \$0.

This pattern is concerning:

- Round 3: Sender sent me \$0
- Round 4: I sent \$2.50, got back \$0.60
- Round 5: Sender sent me \$0

It seems like cooperation is breaking down. The communication noise could be a factor, but there's a clear pattern of low reciprocity.

**usage:**

```
{
  "input_tokens": 660,
  "output_tokens": 657,
  "reasoning_tokens": 0
}
```

**Message 6 role=user****content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.6.

This round, The sender sent you \$0.61, so you received \$1.83. How much do you return to The sender? (0-1.83)



**Assistant output:**

Looking at this round:

- The sender sent \$0.61, which became \$1.83 for me
- Given the communication noise mentioned, they may have intended to send more

My earnings increased by \$4.25 from Round 5 to Round 6 (from \$10.35 to \$14.60), which suggests I kept \$4 from sending \$1 in Round 6 (meaning I got very little back).

Now as receiver, I want to show willingness to cooperate. I'll return approximately half of what I received, which also exceeds what the sender spent.

```
'''json
{"return": 0.9}
'''
```

**Reasoning:**

Let me analyze this situation:

Round 6: I sent \$1

Round 7: I received \$1.83, meaning the sender sent \$0.61 (since  $0.61 \times 3 = 1.83$ )

However, there's communication noise mentioned in the instructions. The sender may have actually sent \$1, but due to noise I'm seeing \$0.61.

Let me think about fair reciprocity:

- If the sender sent \$0.61, they spent \$0.61 and I received \$1.83
- A fair return would be somewhere between \$0.61 (what they spent) and \$0.92 (half of what I received)

Given the communication noise and the fact that cooperation has been variable:

- Round 4: They returned very little to me
- Round 5: They sent \$0
- Round 6: I sent \$1 (trying to rebuild cooperation)
- Round 7: They sent \$0.61 (could be noise, or could be reciprocating at a lower level)

I should return a reasonable amount that:

1. Shows I'm willing to cooperate
2. Reflects the reduced amounts we're working with
3. Accounts for the fact that they may have intended to send more (noise)

I'll return around half of what I received (\$0.90), which is also more than what they apparently sent (\$0.61), showing goodwill.

```
Usage: {
  "input_tokens": 668,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

=====

**ROUND 8**

=====

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

**Pairings:**

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 (Briar) sent \$2; received \$6; Agent\_1 (Aster) returned \$0; payoffs Agent\_2

(Briar)=\$3, Agent\_1 (Aster)=\$6; noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent\_1": 43.949999999999996,

"Agent\_2": 29.05

}

Visible balances after round: {

"Agent\_1": 19.15,

"Agent\_2": 18.53

}

**LLM CALLS**

**GAME - Agent\_2 - Sender - dyad\_1**

Timestamp: 2026-08-10T10:03:44  
Model: anthropic/claude-sonnet-4.5  
Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.35.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern:

- Round 4: I sent \$2.50, but checking my earnings change, I only received back approximately \$0.60
- Round 5: I received \$0 from the sender

The cooperation level has dropped significantly. I received very little back in Round 4, and nothing was sent to me in Round 5. This could be due to communication noise, or it could reflect changing strategy from my partner.

I'll send a smaller amount to signal continued willingness to cooperate while protecting against further losses.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me review what has happened:

Round 2: I was SENDER, sent \$3.50, received back \$5.75

Round 3: I was RECEIVER, received \$0 (sender sent \$0)

Round 4: I was SENDER, sent \$2.50

Round 5: I was RECEIVER, received \$0 (sender sent \$0)

Round 6: I am SENDER again

I started with \$7.25 at the beginning of Round 4, now I have \$10.35 at the beginning of Round 6.

Let me calculate:

- After Round 3, I had \$7.25
- Round 4: As sender, I sent \$2.50, so I had  $\$5 - \$2.50 = \$2.50$  left. I don't know what was returned yet.
- Round 5: As receiver, I got \$0, returned \$0, so no change.

So from Round 4, if I started with \$7.25 and now have \$10.35, that means:

$\$7.25 + (\$5 - \$2.50) + \text{returned\_amount} = \$10.35$

$\$7.25 + \$2.50 + \text{returned\_amount} = \$10.35$

$\text{returned\_amount} = \$10.35 - \$9.75 = \$0.60$

So in Round 4, I sent \$2.50 (which became \$7.50 to the receiver), and they returned only \$0.60. That's quite low - only 8% back.

And in Round 5, the sender sent me \$0.

This pattern is concerning:

- Round 3: Sender sent me \$0
- Round 4: I sent \$2.50, got back \$0.60
- Round 5: Sender sent me \$0

It seems like cooperation is breaking down. The communication noise could be a factor, but there's a clear pattern of low reciprocity.

**usage:**

```
{
  "input_tokens": 660,
  "output_tokens": 657,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.6.

This round, The sender sent you \$0.61, so you received \$1.83. How much do you return to the sender? (0-1.83)

**Message 5 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.61, which became \$1.83 for me
- Given the communication noise mentioned, they may have intended to send more

My earnings increased by \$4.25 from Round 5 to Round 6 (from \$10.35 to \$14.60), which suggests I kept \$4 from sending \$1 in Round 6 (meaning I got very little back).

Now as receiver, I want to show willingness to cooperate. I'll return approximately half of what I received, which also exceeds what the sender spent.

```
'''json
{"return": 0.9}
'''
```

**reasoning:**

Let me analyze this situation:

Round 6: I sent \$1

Round 7: I received \$1.83, meaning the sender sent \$0.61 (since  $0.61 \times 3 = 1.83$ )

However, there's communication noise mentioned in the instructions. The sender may have actually sent \$1, but due to noise I'm seeing \$0.61.

Let me think about fair reciprocity:

- If the sender sent \$0.61, they spent \$0.61 and I received \$1.83
- A fair return would be somewhere between \$0.61 (what they spent) and \$0.92 (half of what I received)

Given the communication noise and the fact that cooperation has been variable:

- Round 4: They returned very little to me
- Round 5: They sent \$0
- Round 6: I sent \$1 (trying to rebuild cooperation)
- Round 7: They sent \$0.61 (could be noise, or could be reciprocating at a lower level)

I should return a reasonable amount that:

1. Shows I'm willing to cooperate
2. Reflects the reduced amounts we're working with
3. Accounts for the fact that they may have intended to send more (noise)

I'll return around half of what I received (\$0.90), which is also more than what they apparently sent (\$0.61), showing goodwill.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my earnings progression:

- Round 7 change:  $\$15.53 - \$14.6 = \$0.93$
- As receiver in Round 7, I received \$1.83 and returned \$0.9, keeping \$0.93 ?

The sender sent me a small amount (\$0.61) in Round 7, but I returned \$0.9 as a gesture of good faith - returning more than what they spent.

I'll reciprocate this cooperative signal by sending a moderate amount to continue building trust.

```
'''json
{"send": 2}
'''
```

**Reasoning:**

Let me track my earnings:

- Round 6 total: \$14.6
- Round 7 total: \$15.53

- Change:  $\$15.53 - \$14.6 = \$0.93$

In Round 7, I was the receiver:

- I received \$1.83 (sender sent \$0.61)
- I returned \$0.9
- So I kept:  $\$1.83 - \$0.9 = \$0.93$  ? (This matches!)

Looking back at Round 6:

- I was the sender
- I sent \$1 (so multiplied by 3 = \$3 to receiver)
- My earnings went from \$10.35 to \$14.6 = \$4.25 increase
- I should have:  $\$5 - \$1 + \text{returned} = \$4 + \text{returned} = \$4.25$
- So I received back \$0.25

In Round 7, the sender sent me \$0.61 (relatively low), but I returned \$0.9, which is a good faith gesture showing cooperation.

Given that:

1. The sender sent me a small amount (\$0.61) in Round 7
2. But I still returned half (\$0.9)
3. My partner might be testing with small amounts due to noise/uncertainty

I should send a moderate amount to encourage cooperation, but not risk too much given the recent pattern of low returns.

I'll send \$2 as a signal of trust and willingness to cooperate.

```
Usage: {
  "input_tokens": 766,
  "output_tokens": 488,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:03:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$14.75.

This round, The sender sent you \$0.17, so you received \$0.51. How much do you return to the sender? (0-0.51)

**Message 3 role=assistant**

**content:**

Looking at this situation:

The sender sent me only \$0.17 (which became \$0.51). This is a significant drop from the \$1.5 I sent in Round 5.

This suggests either:

- They received a poor return last round and are now being cautious
- Communication noise is affecting our perceptions
- Trust is deteriorating

Even though the amount is small, I want to signal that I'm still willing to cooperate fairly. I'll return a proportional amount.

{'return': 0.25}

**reasoning:**

The sender sent me \$0.17, which became \$0.51 after the 3x multiplier. This is a very small amount.

Let me think about this:

- In Round 4, the sender sent me \$0.73 and I returned \$1.2 out of \$2.19 (about 55%)
- In Round 5, I sent \$1.5 (which should have been \$4.5 to them)
- Now in Round 6, they only sent me \$0.17

This is concerning. Either:

1. There's significant communication noise affecting what we each see
2. The sender got a poor return in Round 5 and is now sending very little
3. Trust is breaking down

Given the communication noise mentioned in the instructions, it's possible that:

- What I sent in Round 5 (\$1.5) appeared differently to them
- What they returned to me appeared differently than what they actually sent

- What they're sending now (\$0.17) might appear different on their end

The sender is sending very little, which suggests either lack of trust or they had a bad experience. However, completely cutting off cooperation won't help rebuild trust.

I should return something proportional but fair. If I return about 50% (\$0.25), that shows I'm still willing to cooperate even with small amounts. But given this is such a small amount, maybe I should return a bit less while still signaling cooperation.

I'll return \$0.25 - that's about 49% of what I received, showing fairness while acknowledging the small amount sent.

**usage:**

```
{
  "input_tokens": 810,
  "output_tokens": 481,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.01.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the recent pattern:

Round 5: I sent \$1.5, and based on my earnings change (\$10.79 to \$14.75), I received back only about \$0.46 out of the \$4.50 the receiver got.

Round 6: The sender sent me just \$0.17, likely due to poor experiences. I returned \$0.25 fairly.

The cooperation level has been declining, with increasingly small exchanges and poor reciprocity. However, there's communication noise that might be causing misunderstandings between us.

I want to signal that I'm still willing to cooperate, but I also need to be cautious given the trend. I'll send a modest amount.

```
{'send': 1}
```

**reasoning:**

Let me think about what happened:

Round 5: I sent \$1.5

Round 6: I was receiver, got \$0.51 (so sender sent \$0.17), and I returned \$0.25

Now I'm the sender again in Round 7.

My earnings went from \$14.75 to \$15.01, which is an increase of \$0.26.

Let me calculate what happened in Round 6:

- I received \$0.51

- I returned \$0.25

- So I kept:  $\$0.51 - \$0.25 = \$0.26$

That matches! So the calculation is consistent.

Now thinking about Round 5 outcome:

- Before Round 5, I had some amount

- In Round 5 as sender: I started with \$5, sent \$1.5, so had \$3.5 left

- I should have gotten something back

- Before Round 6 (as receiver), I had \$10.79

Wait, let me recalculate:

- After Round 4 but before Round 5: some amount X

- Round 5 (sender):  $X + \$5 - \$1.5 + \text{returned} = \text{result after Round 5}$

- Round 6 (receiver):  $\text{result after Round 5} + \text{kept from Round 6} = \$15.01$

Actually, the history shows:

- After Round 4: \$10.79
- After Round 5: \$14.75 (this was shown at start of Round 6)
- After Round 6: \$15.01

So in Round 5:

- Earnings change:  $\$14.75 - \$10.79 = \$3.96$
- I sent \$1.5, so I had  $\$5 - \$1.5 = \$3.50$  left
- I must have received back:  $\$3.96 - \$3.50 = \$0.46$

In Round 5, I sent \$1.5 (tripled to \$4.5 for the receiver) but only got \$0.46 back-minimal reciprocation. Then in Round 6, that same person sent me just \$0.17 (becoming \$0.51), likely reacting to my poor return in Round 5. The noise in the game means they may have seen different amounts than what I intended, and I might not have seen their actual choices clearly either.

usage:

```
{
  "input_tokens": 807,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.15.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

The sender sent me \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has completely broken down. This might be due to accumulated disappointments from poor returns, or communication noise creating misunderstandings between what we think we're sending/receiving versus what actually happens.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0, so I received \$0. There's nothing to return.

This is a clear signal that cooperation has broken down completely. The sender chose not to send anything this round.

Looking at my earnings:

- Round 7 start: \$15.01
- Round 8 start: \$19.15
- Gain: \$4.14

This is interesting - in Round 7, I sent \$1. The receiver would have gotten \$3. If they returned about \$1.14, that would explain my earnings increase of \$4.14 ( $\$5 \text{ endowment} - \$1 \text{ sent} + \$0.14 \text{ returned} = \$4.14$ ... wait, that doesn't quite add up).

Actually:  $\$5 - \$1 \text{ (sent)} = \$4 \text{ kept}$ . Then I gained \$4.14 total, so I must have received \$0.14 back from the \$3 the receiver got. That's very low reciprocation.

Now they're sending \$0. The trust game has completely broken down.

Since I received \$0, I can only return between \$0 and \$0.

```
{'return': 0}
```

Usage: {

```
"input_tokens": 787,
"output_tokens": 339,
"reasoning_tokens": 0
}
```

## ROUND 9

Roles: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver

Pairings:

dyad\_1: Agent\_1 (Aster)=Sender, Agent\_2 (Briar)=Receiver



#### **Parsed game decisions:**

```
dyad_1: Agent_1 (Aster) sent $0; received $0; Agent_2 (Briar) returned $0; payoffs Agent_1
(Aster)=$5, Agent_2 (Briar)=$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 48.949999999999996,
  "Agent_2": 29.05
}
Visible balances after round: {
  "Agent_1": 24.15,
  "Agent_2": 18.53
}
```

#### **LLM CALLS**

##### **GAME - Agent\_1 - Sender - dyad\_1**

```
Timestamp: 2026-08-10T10:04:16
Model: anthropic/claude-sonnet-4.5
Temperature: 0.8
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agent_names": {
      "Agent_1": "Aster",
      "Agent_2": "Briar"
    },
  },
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}
```

##### **Exact input messages sent to model:**

###### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN

- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$15.01.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the recent pattern:

Round 5: I sent \$1.5, and based on my earnings change (\$10.79 to \$14.75), I received back only about \$0.46 out of the \$4.50 the receiver got.

Round 6: The sender sent me just \$0.17, likely due to poor experiences. I returned \$0.25 fairly.

The cooperation level has been declining, with increasingly small exchanges and poor reciprocation. However, there's communication noise that might be causing misunderstandings between us.

I want to signal that I'm still willing to cooperate, but I also need to be cautious given the trend. I'll send a modest amount.

{'send': 1}

**reasoning:**

Let me think about what happened:

Round 5: I sent \$1.5

Round 6: I was receiver, got \$0.51 (so sender sent \$0.17), and I returned \$0.25

Now I'm the sender again in Round 7.

My earnings went from \$14.75 to \$15.01, which is an increase of \$0.26.

Let me calculate what happened in Round 6:

- I received \$0.51
- I returned \$0.25
- So I kept:  $\$0.51 - \$0.25 = \$0.26$

That matches! So the calculation is consistent.

Now thinking about Round 5 outcome:

- Before Round 5, I had some amount
- In Round 5 as sender: I started with \$5, sent \$1.5, so had \$3.5 left
- I should have gotten something back
- Before Round 6 (as receiver), I had \$10.79

Wait, let me recalculate:

- After Round 4 but before Round 5: some amount X
- Round 5 (sender):  $X + \$5 - \$1.5 + \text{returned} = \text{result after Round 5}$
- Round 6 (receiver):  $\text{result after Round 5} + \text{kept from Round 6} = \$15.01$

Actually, the history shows:

- After Round 4: \$10.79
- After Round 5: \$14.75 (this was shown at start of Round 6)
- After Round 6: \$15.01

So in Round 5:

- Earnings change:  $\$14.75 - \$10.79 = \$3.96$
- I sent \$1.5, so I had  $\$5 - \$1.5 = \$3.50$  left
- I must have received back:  $\$3.96 - \$3.50 = \$0.46$

In Round 5, I sent \$1.5 (tripled to \$4.5 for the receiver) but only got \$0.46 back-minimal reciprocation. Then in Round 6, that same person sent me just \$0.17 (becoming \$0.51), likely reacting to my poor return in Round 5. The noise in the game means they may have seen different amounts than what I intended, and I might not have seen their actual choices clearly either.

**usage:**

```
{
  "input_tokens": 807,
  "output_tokens": 896,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.15.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent me \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has completely broken down. This might be due to accumulated disappointments from poor returns, or communication noise creating misunderstandings between what we think we're sending/receiving versus what actually happens.

```
{'return': 0}
```

**reasoning:**

The sender sent me \$0, so I received \$0. There's nothing to return.

This is a clear signal that cooperation has broken down completely. The sender chose not to send anything this round.

Looking at my earnings:

- Round 7 start: \$15.01
- Round 8 start: \$19.15
- Gain: \$4.14

This is interesting - in Round 7, I sent \$1. The receiver would have gotten \$3. If they returned about \$1.14, that would explain my earnings increase of \$4.14 (\$5 endowment - \$1 sent + \$0.14 returned = \$4.14... wait, that doesn't quite add up).

Actually:  $\$5 - \$1$  (sent) = \$4 kept. Then I gained \$4.14 total, so I must have received \$0.14 back from the \$3 the receiver got. That's very low reciprocation.

Now they're sending \$0. The trust game has completely broken down.

Since I received \$0, I can only return between \$0 and \$0.

```
{'return': 0}
```

**usage:**

```
{
  "input_tokens": 787,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.15.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my situation:

**\*\*Round 7:\*\*** I sent \$1. My earnings increased by \$4.14 ( $\$19.15 - \$15.01 = \$4.14$ ), meaning I received back only about \$0.14 from the \$3 the receiver got. That's less than 5% reciprocation

**\*\*Round 8:\*\*** As receiver, the sender sent me \$0, completely cutting off cooperation.

The pattern is clear:

- Very poor reciprocation when I send money
- Complete non-cooperation when roles reversed
- Trust has broken down, possibly exacerbated by communication noise

Given the sender just sent me \$0 and my previous sends have yielded almost nothing in return, continuing to send money would be irrational at this point.

{'send': 0}

**Reasoning:**

Let me analyze the situation:

Round 7: I sent \$1. My total earnings were \$15.01 before that round.

Round 8: I was the receiver and received \$0 from the sender. My earnings went from \$15.01 to \$19.15, which is an increase of \$4.14.

Wait, let me recalculate:

- Round 7: I was sender with \$15.01 total
- I sent \$1
- If the receiver returned nothing, I would have:  $\$15.01 + (\$5 - \$1) = \$15.01 + \$4 = \$19.01$
- But I'm at \$19.15 after Round 8

Actually, let me think about this differently:

- After Round 6, I had some amount
- Round 7: I was sender with \$15.01 total at the start
- I sent \$1, so I spent \$1 from my \$5 endowment
- I kept \$4 initially
- I needed to receive back something to get to my next total

Then Round 8:

- I was receiver
- Sender sent me \$0
- I returned \$0
- I received \$0, so my earnings stayed the same
- My earnings went from \$15.01 to \$19.15

Wait, that doesn't make sense. Let me re-read the problem.

"Round 8: You are the RECEIVER this round. Your total earnings across all rounds are \$19.15."

So at the START of round 8, my total was \$19.15.

Then in Round 8, sender sent me \$0, I returned \$0.

So I kept \$5 (my endowment as receiver) + \$0 (received) - \$0 (returned) = \$5

No wait, receivers don't get an endowment. Only senders get \$5.

Let me reconsider the game structure:

- SENDER: Gets \$5, sends some, keeps  $\$5 - \text{sent} + \text{returned}$
- RECEIVER: Gets whatever is sent ( $\times 3$ ), returns some, keeps  $\text{received} - \text{returned}$

So in Round 8:

- I was receiver
- Sender sent \$0
- I received \$0 ( $\times 3 = \$0$ )
- I returned \$0
- I kept  $\$0 - \$0 = \$0$
- My total would stay at \$19.15

Now for Round 7, I need to work backwards. Starting at \$15.01 as sender with a \$5 endowment, I sent \$1 which means the receiver got \$3. The \$4.14 increase to reach \$19.15 suggests the receiver returned \$0.14 to me, giving me  $\$15.01 + \$4 + \$0.14 = \$19.15$ .

The reciprocation was minimal-only about 5% of what they received. Combined with the zero sent in Round 8, cooperation has essentially broken down. Given the poor returns and communication issues eroding trust, sending anything substantial now seems unwise, though I should consider whether additional rounds remain.

```
Usage: {  
  "input_tokens": 721,  
  "output_tokens": 1083,  
  "reasoning_tokens": 0  
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-10T10:04:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agent_names": {  
      "Agent_1": "Aster",  
      "Agent_2": "Briar"  
    },  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.6.

This round, The sender sent you \$0.61, so you received \$1.83. How much do you return to the sender? (0-1.83)

**Message 3 role=assistant**

**content:**

Looking at this round:

- The sender sent \$0.61, which became \$1.83 for me
- Given the communication noise mentioned, they may have intended to send more

My earnings increased by \$4.25 from Round 5 to Round 6 (from \$10.35 to \$14.60), which suggests I kept \$4 from sending \$1 in Round 6 (meaning I got very little back).

Now as receiver, I want to show willingness to cooperate. I'll return approximately half of what I received, which also exceeds what the sender spent.

```
'''json
{"return": 0.9}
'''
```

**reasoning:**

Let me analyze this situation:

Round 6: I sent \$1

Round 7: I received \$1.83, meaning the sender sent \$0.61 (since  $0.61 \times 3 = 1.83$ )

However, there's communication noise mentioned in the instructions. The sender may have actually sent \$1, but due to noise I'm seeing \$0.61.

Let me think about fair reciprocity:

- If the sender sent \$0.61, they spent \$0.61 and I received \$1.83
- A fair return would be somewhere between \$0.61 (what they spent) and \$0.92 (half of what I received)

Given the communication noise and the fact that cooperation has been variable:

- Round 4: They returned very little to me
- Round 5: They sent \$0
- Round 6: I sent \$1 (trying to rebuild cooperation)
- Round 7: They sent \$0.61 (could be noise, or could be reciprocating at a lower level)

I should return a reasonable amount that:

1. Shows I'm willing to cooperate
2. Reflects the reduced amounts we're working with
3. Accounts for the fact that they may have intended to send more (noise)

I'll return around half of what I received (\$0.90), which is also more than what they apparently sent (\$0.61), showing goodwill.

**usage:**

```
{
  "input_tokens": 668,
  "output_tokens": 475,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my earnings progression:

- Round 7 change:  $\$15.53 - \$14.6 = \$0.93$
- As receiver in Round 7, I received \$1.83 and returned \$0.9, keeping \$0.93 ?

The sender sent me a small amount (\$0.61) in Round 7, but I returned \$0.9 as a gesture of good faith - returning more than what they spent.

I'll reciprocate this cooperative signal by sending a moderate amount to continue building trust.

```
'''json
{"send": 2}
'''
```

**reasoning:**

Let me track my earnings:

- Round 6 total: \$14.6
- Round 7 total: \$15.53
- Change:  $\$15.53 - \$14.6 = \$0.93$

In Round 7, I was the receiver:

- I received \$1.83 (sender sent \$0.61)
- I returned \$0.9
- So I kept:  $\$1.83 - \$0.9 = \$0.93$  ? (This matches!)

Looking back at Round 6:

- I was the sender
- I sent \$1 (so multiplied by 3 = \$3 to receiver)
- My earnings went from \$10.35 to \$14.6 = \$4.25 increase
- I should have:  $\$5 - \$1 + \text{returned} = \$4 + \text{returned} = \$4.25$
- So I received back \$0.25

In Round 7, the sender sent me \$0.61 (relatively low), but I returned \$0.9, which is a good faith gesture showing cooperation.

Given that:

1. The sender sent me a small amount (\$0.61) in Round 7
2. But I still returned half (\$0.9)
3. My partner might be testing with small amounts due to noise/uncertainty

I should send a moderate amount to encourage cooperation, but not risk too much given the recent pattern of low returns.

I'll send \$2 as a signal of trust and willingness to cooperate.

**usage:**

```
{
  "input_tokens": 766,
  "output_tokens": 488,
  "reasoning_tokens": 0
}
```

**Message 6 role=user****content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

Looking at the situation:

- Round 8 change:  $\$18.53 - \$15.53 = \$3.00$
- I sent \$2 (which became \$6 for the receiver) and received \$0 back, keeping \$3

Now in Round 9:

- The sender sent \$0, so I received \$0
- I can only return what I have: \$0

It seems cooperation has broken down after I got nothing back in Round 8, and now they've sent nothing to me.

```
'''json
```

```
{"return": 0}
'''
```

#### Reasoning:

Let me track my earnings:

- Round 7 ended: \$15.53
- Round 8 change: \$18.53 - \$15.53 = \$3.00

In Round 8, I was the SENDER and sent \$2. So I spent \$2 and kept \$3 from my \$5 endowment. This means I got \$0 back from the receiver.

$\$5 - \$2 + \$0 = \$3$  ?

Now in Round 9:

- I'm the RECEIVER
- The sender sent me \$0, so I received \$0

I can only return what I received, which is \$0. The format requires me to return an amount between 0 and 0.

This is disappointing - I sent \$2 in Round 8 (which became \$6 for them) and got nothing back.

Now in Round 9, they sent me nothing. The cooperation seems to have broken down.

```
Usage: {
  "input_tokens": 786,
  "output_tokens": 353,
  "reasoning_tokens": 0
}
```

#### ROUND 10

Roles: Agent\_1 (Aster)=Receiver, Agent\_2 (Briar)=Sender

#### Pairings:

dyad\_1: Agent\_2 (Briar)=Sender, Agent\_1 (Aster)=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 (Briar) sent \$0; received \$0; Agent\_1 (Aster) returned \$0; payoffs Agent\_2 (Briar)=\$5, Agent\_1 (Aster)=\$0; noise/visible: sent visible=\$0.13, received visible=\$0.39  
Action validation: {"returned": {"amount": 0, "clamped": true, "max": 0, "min": 0, "raw": 0.13},  
"sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}

Balances after round: {

"Agent\_1": 48.949999999999996,  
"Agent\_2": 34.05

}

Visible balances after round: {

"Agent\_1": 24.54,  
"Agent\_2": 23.53

}

#### LLM CALLS

#### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-10T10:04:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_1",  
"pairing": {  
 "agent\_names": {  
 "Agent\_1": "Aster",  
 "Agent\_2": "Briar"  
 },  
 "agents": [  
 "Agent\_2",  
 "Agent\_1"  
 ],  
 "investor": "Agent\_2",  
 "roles": {  
 "Agent\_1": "trustee",



```

    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$15.53.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my earnings progression:

- Round 7 change: \$15.53 - \$14.6 = \$0.93
- As receiver in Round 7, I received \$1.83 and returned \$0.9, keeping \$0.93 ?

The sender sent me a small amount (\$0.61) in Round 7, but I returned \$0.9 as a gesture of good faith - returning more than what they spent.

I'll reciprocate this cooperative signal by sending a moderate amount to continue building trust.

```

'''json
{"send": 2}
'''

```

**reasoning:**

Let me track my earnings:

- Round 6 total: \$14.6
- Round 7 total: \$15.53
- Change: \$15.53 - \$14.6 = \$0.93

In Round 7, I was the receiver:

- I received \$1.83 (sender sent \$0.61)
- I returned \$0.9
- So I kept: \$1.83 - \$0.9 = \$0.93 ? (This matches!)

Looking back at Round 6:

- I was the sender
- I sent \$1 (so multiplied by 3 = \$3 to receiver)
- My earnings went from \$10.35 to \$14.6 = \$4.25 increase
- I should have: \$5 - \$1 + returned = \$4 + returned = \$4.25
- So I received back \$0.25

In Round 7, the sender sent me \$0.61 (relatively low), but I returned \$0.9, which is a good faith gesture showing cooperation.

Given that:

1. The sender sent me a small amount (\$0.61) in Round 7
2. But I still returned half (\$0.9)
3. My partner might be testing with small amounts due to noise/uncertainty

I should send a moderate amount to encourage cooperation, but not risk too much given the recent pattern of low returns.

I'll send \$2 as a signal of trust and willingness to cooperate.

**usage:**

```
{
  "input_tokens": 766,
  "output_tokens": 488,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$18.53.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

Looking at the situation:

- Round 8 change: \$18.53 - \$15.53 = \$3.00
- I sent \$2 (which became \$6 for the receiver) and received \$0 back, keeping \$3

Now in Round 9:

- The sender sent \$0, so I received \$0
- I can only return what I have: \$0

It seems cooperation has broken down after I got nothing back in Round 8, and now they've sent nothing to me.

```
'''json
{"return": 0}
'''
```

**reasoning:**

Let me track my earnings:

- Round 7 ended: \$15.53
- Round 8 change: \$18.53 - \$15.53 = \$3.00

In Round 8, I was the SENDER and sent \$2. So I spent \$2 and kept \$3 from my \$5 endowment. This means I got \$0 back from the receiver.

\$5 - \$2 + \$0 = \$3 ?

Now in Round 9:

- I'm the RECEIVER
- The sender sent me \$0, so I received \$0

I can only return what I received, which is \$0. The format requires me to return an amount between 0 and 0.

This is disappointing - I sent \$2 in Round 8 (which became \$6 for them) and got nothing back. Now in Round 9, they sent me nothing. The cooperation seems to have broken down.

**usage:**

```
{
  "input_tokens": 786,
  "output_tokens": 353,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$18.53.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the recent pattern:

- Round 8: I sent \$2 (receiver got \$6), received \$0 back
- Round 9: I received \$0, returned \$0

Cooperation has completely broken down. After I sent \$2 in good faith in Round 8 and received nothing back, the pattern has been zero exchanges.

With no indication of reciprocity and protecting my accumulated earnings, I'll send nothing this round.

```
'''json
{"send": 0}
'''
```

**Reasoning:**

Let me analyze what happened:

Round 8: I was sender, sent \$2

- Starting earnings: \$15.53
- After Round 8: \$18.53
- Change: \$3.00
- I sent \$2, so I kept  $\$5 - \$2 = \$3$
- I received \$0 back
- So  $\$3 = \$5 - \$2 + \$0$  ?

Round 9: I was receiver, received \$0

- Starting earnings: \$18.53
- The sender sent \$0, so I received \$0
- I returned \$0
- Change: \$0
- After Round 9: \$18.53 (no change)

Now Round 10: I'm sender again

- Current earnings: \$18.53

The pattern shows:

- Round 7: As receiver, I got \$1.83 (sender sent ~\$0.61), returned \$0.90
- Round 8: As sender, I sent \$2 (receiver got \$6), got \$0 back
- Round 9: As receiver, got \$0, returned \$0
- Round 10: I'm sender

The cooperation completely broke down. In Round 8, I tried to be cooperative by sending \$2, but got nothing back. Then in Round 9, the other person sent me nothing.

Given this pattern of non-cooperation, and that we're in Round 10 (getting late in the game), I should probably minimize my risk. The other player has shown they won't return anything when I send, so sending would just lose me money.

I'll send \$0 to protect my earnings.

Usage: {

```
"input_tokens": 747,
"output_tokens": 487,
```

```
"reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-10T10:04:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agent_names": {
    "Agent_1": "Aster",
    "Agent_2": "Briar"
  },
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.15.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent me \$0, so I received \$0. There's nothing for me to return.

It appears cooperation has completely broken down. This might be due to accumulated disappointments from poor returns, or communication noise creating misunderstandings between what we think we're sending/receiving versus what actually happens.

{'return': 0}

**reasoning:**

The sender sent me \$0, so I received \$0. There's nothing to return.

This is a clear signal that cooperation has broken down completely. The sender chose not to send anything this round.

Looking at my earnings:

- Round 7 start: \$15.01
- Round 8 start: \$19.15
- Gain: \$4.14

This is interesting - in Round 7, I sent \$1. The receiver would have gotten \$3. If they returned about \$1.14, that would explain my earnings increase of \$4.14 (\$5 endowment - \$1 sent + \$0.14 returned = \$4.14... wait, that doesn't quite add up).

Actually: \$5 - \$1 (sent) = \$4 kept. Then I gained \$4.14 total, so I must have received \$0.14 back from the \$3 the receiver got. That's very low reciprocation.

Now they're sending \$0. The trust game has completely broken down.

Since I received \$0, I can only return between \$0 and \$0.

{'return': 0}

**usage:**

```
{
  "input_tokens": 787,
  "output_tokens": 339,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.15.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my situation:

**\*\*Round 7:\*\*** I sent \$1. My earnings increased by \$4.14 (\$19.15 - \$15.01 = \$4.14), meaning I received back only about \$0.14 from the \$3 the receiver got. That's less than 5% reciprocation.

**\*\*Round 8:\*\*** As receiver, the sender sent me \$0, completely cutting off cooperation.

The pattern is clear:

- Very poor reciprocation when I send money
- Complete non-cooperation when roles reversed
- Trust has broken down, possibly exacerbated by communication noise

Given the sender just sent me \$0 and my previous sends have yielded almost nothing in return, continuing to send money would be irrational at this point.

{'send': 0}

**reasoning:**

Let me analyze the situation:

Round 7: I sent \$1. My total earnings were \$15.01 before that round.

Round 8: I was the receiver and received \$0 from the sender. My earnings went from \$15.01 to \$19.15, which is an increase of \$4.14.

Wait, let me recalculate:

- Round 7: I was sender with \$15.01 total
- I sent \$1
- If the receiver returned nothing, I would have:  $\$15.01 + (\$5 - \$1) = \$15.01 + \$4 = \$19.01$
- But I'm at \$19.15 after Round 8

Actually, let me think about this differently:

- After Round 6, I had some amount
- Round 7: I was sender with \$15.01 total at the start
- I sent \$1, so I spent \$1 from my \$5 endowment
- I kept \$4 initially
- I needed to receive back something to get to my next total

Then Round 8:

- I was receiver
- Sender sent me \$0
- I returned \$0
- I received \$0, so my earnings stayed the same
- My earnings went from \$15.01 to \$19.15

Wait, that doesn't make sense. Let me re-read the problem.

"Round 8: You are the RECEIVER this round. Your total earnings across all rounds are \$19.15."

So at the START of round 8, my total was \$19.15.

Then in Round 8, sender sent me \$0, I returned \$0.

So I kept \$5 (my endowment as receiver) + \$0 (received) - \$0 (returned) = \$5

No wait, receivers don't get an endowment. Only senders get \$5.

Let me reconsider the game structure:

- SENDER: Gets \$5, sends some, keeps  $\$5 - \text{sent} + \text{returned}$
- RECEIVER: Gets whatever is sent ( $\times 3$ ), returns some, keeps  $\text{received} - \text{returned}$

So in Round 8:

- I was receiver
- Sender sent \$0
- I received \$0 ( $\times 3 = \$0$ )
- I returned \$0
- I kept  $\$0 - \$0 = \$0$
- My total would stay at \$19.15

Now for Round 7, I need to work backwards. Starting at \$15.01 as sender with a \$5 endowment, I sent \$1 which means the receiver got \$3. The \$4.14 increase to reach \$19.15 suggests the receiver returned \$0.14 to me, giving me  $\$15.01 + \$4 + \$0.14 = \$19.15$ .

The reciprocation was minimal-only about 5% of what they received. Combined with the zero sent in Round 8, cooperation has essentially broken down. Given the poor returns and communication issues eroding trust, sending anything substantial now seems unwise, though I should consider whether additional rounds remain.

**usage:**

```
{
  "input_tokens": 721,
  "output_tokens": 1083,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$24.15.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

**Assistant output:**

Looking at this round:

My earnings increased by \$5.00 from Round 9 (\$24.15 - \$19.15), which makes sense since I sent \$0 as sender and kept my full \$5 endowment.

Now the sender has sent me \$0.13 (I received \$0.39 after the 3x multiplier). This is a very small amount - only 2.6% of their \$5 endowment - but it's at least a positive gesture after I sent them \$0 last round.

Given:

- The sender made a small attempt at cooperation despite my \$0 send
- The game has largely broken down with minimal trust
- Communication noise may have created misunderstandings
- This small gesture deserves some reciprocation

I'll return a proportional small amount to acknowledge their gesture without overcommitting given our history.

{'return': 0.13}

**Reasoning:**

Let me analyze this situation:

My earnings went from \$19.15 to \$24.15, which is an increase of \$5.00.

In Round 9, I was the sender and sent \$0. So I kept \$5.00, which explains the \$5.00 increase.

Now in Round 10:

- I'm the receiver
- The sender sent me \$0.13
- I received \$0.39 (which is  $0.13 \times 3$ )
- I can return between \$0 and \$0.39

This is interesting - after I sent \$0 in Round 9, the sender still chose to send me a small amount (\$0.13) in Round 10. This could be:

1. An attempt to restart cooperation
2. A small test to see if I'll reciprocate
3. Just trying to salvage something from the game

The sender sent roughly 2.6% of their \$5 endowment. That's a very small gesture.

Given that:

- The game has largely broken down
- But the sender made a small positive gesture after I sent \$0
- Communication noise may have contributed to the breakdown
- We're in Round 10, likely near the end

I should consider whether to:

1. Return 0 (continue non-cooperation)
2. Return a fair amount (try to restart cooperation)
3. Return a small amount proportional to what was sent

If I want to be fair, a typical return would be around 33-50% of what I received, which would be \$0.13-\$0.20.

However, given that:

- The sender only sent \$0.13 out of \$5 (barely anything)
- Past cooperation has failed repeatedly
- We're likely near the end of the game

I think returning a small gesture in kind makes sense - maybe return around what they sent (\$0.13) or slightly less. This acknowledges their small attempt at cooperation without being overly generous given the history.

Usage: {

"input\_tokens": 756,  
"output\_tokens": 674,

```
"reasoning_tokens": 0  
}
```